

Steering Idioms: Controlling Figurative-vs-Literal Interpretation via Residual-Stream Activation Steering

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Abstract

Large language models (LLMs) default to figurative interpretations of idiomatic expressions when context supports them. We investigate whether this bias can be directly controlled via activation steering: asking whether idiomaticity is encoded as a manipulable direction in residual space. We introduce IdiomSteer, a small controlled benchmark of 100 sentences across 20 idioms designed to expose interpretation-dependent generation. Using a steering vector applied to five LLMs from two model families, ranging from 2B to 12B parameters, we show that steering systematically shifts model outputs toward literal interpretations while preserving fluency. Three findings stand out: cross-layer steering—vectors from mid-to-late layers injected at early layers—consistently outperforms same-layer interventions; steering vectors transfer to held-out idioms, suggesting a general figurative-vs-literal direction in residual space; and effects replicate across architectures and model sizes using the same intervention protocol.

1 Introduction

Natural language frequently exhibits non-compositional semantic constructions, where the meaning of an expression cannot be derived directly from the meanings of its individual components. A central instance of this phenomenon is *idiomatic expressions*, whose interpretation depends on convention and context rather than compositional semantics. For example, the phrase “*bad apple*” can denote either a literal defective fruit or, more commonly, a figurative reference to a corrupt or problematic individual. Resolving such ambiguity requires contextual reasoning, making idioms a core challenge in semantic modeling.

Recent work has shown that large language models (LLMs) capture aspects of non-compositional meaning with increasing robustness. In particular, Hashiloni et al. (2025) and Hefetz et al. (2025)

Prefix: During the party, Sarah tried to *break the ice*

Unsteered: During the party, Sarah tried to *break the ice* with the other guests by talking about her job.

Steered: During the party, Sarah tried to *break the ice* into pieces so that the ice cubes would melt.

Figure 1: Example of a prefix and model continuation with and without steering toward literal interpretation. The unsteered continuation follows the conventional figurative reading of *break the ice*, while the steered continuation uses the idiom’s component words compositionally.

demonstrate that LLMs can reliably identify idiomatic expressions and distinguish them from literal usages across contexts, including in cross-lingual settings. Earlier work shows that while LLMs achieve strong performance on idiom detection, they still exhibit systematic biases and limitations. For example, De Luca Fornaciari et al. (2024) introduce a challenging benchmark for idiom detection and show that performance degrades under more complex contextual conditions, while Phelps et al. (2024) find that even the largest LLMs fall short of fine-tuned models on idiomaticity detection. Similarly, Khoshtab et al. (2025) highlight variability in idiom understanding across languages and figurative constructions. Together, these findings indicate that LLMs encode internal representations that support idiomaticity, but that these representations also reflect strong default preferences toward figurative interpretation.

A complementary line of work studies how internal representations of high-level model behavior can be *controlled* at inference time through residual-stream activation steering (Turner et al., 2024; Rimskey et al., 2024; Arditi et al., 2024). By identifying a vector in residual space and injecting it during generation, these methods causally bias model behavior in semantically meaningful

ways. Most prior work in this area targets relatively low-level properties (sentiment, refusal, factual associations); the application of activation steering to higher-order semantic phenomena such as figurative-vs-literal interpretation has received comparatively little attention.

We investigate whether idiomaticity can be captured as a dense vector that directly controls the figurative-vs-literal balance at inference time. Using a white-box steering approach, we construct an idiomaticity vector from training-time activations and inject it during generation. Without intervention, base models default to figurative continuations; with steering, outputs shift systematically toward literal interpretations. Evaluated via human annotations and LLM-as-a-judge across models of varying architectures and sizes, our results indicate that non-compositional semantic behavior can be captured and manipulated through vector-based interventions in representation space. Figure 1 illustrates this effect qualitatively: the unsteered model follows the conventional figurative reading of *break the ice*, while steering in the literal direction elicits a literal continuation about ice as a physical object. See more examples in Appendix A.

We make the following contributions:

1. We introduce **IdiomSteer**, a controlled benchmark of 100 sentence prefixes across 20 English idioms, each ending in an idiomatic expression that admits both figurative and literal continuations.
2. We show that activation steering reliably shifts LLM continuations toward literal interpretation across five base models from two architectural families, ranging from 2B to 12B parameters, while maintaining generation coherence. We provide a comparative analysis demonstrating consistent steering effects across architectures and sizes.
3. We analyze source-layer and injection-layer choices, and show that the steering vector is not tied to the particular idioms used to construct it: vectors built from different training idiom sets are geometrically aligned and produce consistent effects on held-out idioms.

To guide our investigation, we address the following research questions:

RQ1. Can residual-stream activation steering reliably control whether an LLM produces a figurative or literal continuation of an idiomatic expression, across models of varying architecture and scale?

RQ2. Does the resulting steering vector reflect a

directional intervention rather than generic activation perturbation?

RQ3. What layer-choice regime makes steering effective?

RQ4. Does the steering vector capture an idiom-general property, rather than depending on the specific idioms used for vector construction or evaluation?

The IdiomSteer dataset, construction guidelines, annotation guidelines, and evaluation resources will be released in the camera-ready version.

2 Related Work

2.1 Idioms in LLMs

Idiomatic expressions are a central case of non-compositional semantics, where meaning cannot be derived directly from the constituent words (Baldwin and Kim, 2010; Constant et al., 2017). Prior work has largely studied idioms through *identification and classification*, that is, determining whether an expression is used figuratively or literally in context. Datasets such as MAGPIE (Haagsma et al., 2020), MIDAS (Kim et al., 2025), and related benchmarks (Mi et al., 2025) have enabled systematic evaluation of this ability. Recent studies show that prompted LLMs can achieve competitive performance on standard idiom detection benchmarks (Hashiloni et al., 2025; Arslan et al., 2025), though performance remains sensitive to contextual complexity and language variation (Phelps et al., 2024; De Luca Fornaciari et al., 2024; Khoshtab et al., 2025).

A complementary line of work examines how idiomaticity is represented internally in LLMs. Hefetz et al. (2025) use probing and Logit Lens (nostalgebraist, 2020) to show that idiom-vs-literal information is partially recoverable from intermediate layers; Oh et al. (2026) use causal tracing to characterize a three-stage mechanism in which early layers retrieve figurative interpretations while parallel pathways carry the literal compositional reading forward.

Most relevantly, Ghosh and Jiang (2026) report that LLMs compress concreteness into a one-dimensional direction in mid-to-late layers and use this direction to bias generation toward more literal or figurative text. Our work shares this motivation but differs in focus: we target interpretation selection rather than generation style, operating at the level of idiomatic expressions, multi-word units whose meaning is not derivable from their compo-

nents, where the task is to resolve a phrasal ambiguity rather than a lexical one. Additionally, we identify a cross-layer steering regime, separating the layer at which the direction is constructed from the layer at which it is injected, which same-layer approaches do not exploit.

2.2 Activation Steering for Inference Time

Activation steering controls model behavior by intervening on the residual stream of a frozen language model during inference. Turner et al. (2024) introduce activation addition, deriving steering directions from activation differences between contrasting prompts; Rimsky et al. (2024) formalize this as Contrastive Activation Addition (CAA), using differences between class-conditional activation means—the construction we follow in this work for our main steering vector. Some high-level behaviors are mediated by low-dimensional directions in residual space, including a refusal direction (Arditi et al., 2024) and a sentiment direction (Tigges et al., 2024); we compare against a probe-derived direction following this line of work, and evaluate an angular operator inspired by Vu and Nguyen (2025), which steers by rotating activations rather than adding a scaled vector.

Most prior steering work targets surface-level or behaviorally coarse properties (e.g., sentiment, refusal, truthfulness, writing style, factual associations). We extend this line of research to a higher-order linguistic-semantic phenomenon: the figurative-vs-literal interpretation of idiomatic expressions, asking whether activation steering can control non-compositional meaning.

3 IdiomSteer Benchmark

To study inference-time control over figurative-vs-literal interpretation, we introduce **IdiomSteer**, a controlled benchmark of 100 sentence prefixes spanning 20 English idioms (five variants per idiom) drawn from MAGPIE (Haagsma et al., 2020) and validated by two independent annotators (Appendix B).

Design. Each sentence is constructed so that the target idiom appears as the final tokens of the sentence—with no post-idiom context—and the pre-idiom context is deliberately written to leave the figurative-vs-literal ambiguity unresolved. For example, “During the party, Sarah tried to *break the ice*” admits both a figurative continuation (e.g., ... *by talking to the man next to her*) and a literal

one (e.g., ... *into pieces so that the ice cubes would melt*). The two readings need not be equally likely; the criterion is that both continuations are natural. This bidirectional plausibility ensures that any failure to produce a target interpretation reflects a property of the model or the steering method, not a benchmark artifact in which one direction was never available.

Construction and validation. We use Claude 3.5 Sonnet (Anthropic, 2024) only to propose candidate sentences during drafting. All final examples are curated, edited, and validated by human annotators. Two independent human annotators then validate each sentence under a dual-completion protocol: both annotators produce a figurative and a literal continuation; a sentence is *valid* only if both continuations are natural. Sentences flagged by either annotator are revised and re-circulated until all 100 are *valid*. The 20 IdiomSteer idioms are entirely distinct from those used to estimate steering vectors. Additional details on annotators, construction constraints, validation, and iterative refinement are provided in Appendix B.

4 Steering Methodology

We use residual stream activation steering (Turner et al., 2024; Rimsky et al., 2024) to bias model continuations toward a literal or figurative reading. Our pipeline has three components: (i) the *steering vector*, a unit-norm direction in residual space; (ii) the *steering operator*, combining the vector with an activation; and (iii) the *application protocol*, specifying where and when the operator fires.

4.1 Steering Vector Construction

We compare three constructions of a steering vector $v \in R^d$, where d is the residual-stream dimensionality (e.g., $d = 3,072$ for Llama-3.2-3B). All vectors are unit-normalized, so intervention magnitude is controlled entirely by the scaling coefficient α (Section 4.2) and is comparable across vectors and models. Each vector is computed once from a fixed training pool and reused at inference.

Activation collection. All three vectors are derived from *residual-stream activations*: the per-layer, per-token hidden states that pass between transformer blocks. For each training sentence, we run a forward pass through the frozen model and record the residual-stream activation at the final token of the idiomatic expression, at a designated source layer L_s . Note that L_s is distinct from the

layer at which the vector is later injected during steering (Section 4.3). Each recorded activation is a vector in R^d .

Training pool. We construct steering vectors from IdioLink (Hashiloni et al., 2026), a benchmark of sentences spanning both figurative and literal uses of English idioms. We use the idiom-disjoint train split, retaining 1,683 sentences across 22 idioms after filtering for surface-form idiom presence (~ 76 sentences per idiom; 629 figurative, 1,054 literal). Each sentence is annotated as either *figurative* (the idiom carries its conventional non-compositional meaning) or *literal* (the component words are used compositionally outside the idiomatic frame). The idiomatic expression is explicitly marked in each sentence, making it straightforward to identify its constituent tokens. We denote the resulting activation sets at the source layer L_s as H^{fig} and H^{lit} . The class distribution is moderately imbalanced ($\sim 37\%$ figurative / $\sim 63\%$ literal). Additional details are provided in Appendix C.

We experiment with the three vector construction methods, which we describe in the rest of this section: Mean-difference (MD), a contrastive average between class centroids; Probe (PROBE), the weight vector of a learned classifier; and Random (RAND), an untrained perturbation control.

Mean-difference (MD). Following the standard construction of contrastive-activation (Rimsky et al., 2024; Arditì et al., 2024), we compute the difference between the class-conditional means of the figurative and literal activation sets, and unit-normalize. Let $c \in \{\text{fig}, \text{lit}\}$ index the two classes, and let $\bar{h}^c$ denote the empirical mean activation of class c :

$$\bar{h}^c = \frac{1}{|H^c|} \sum_{h \in H^c} h, \quad v_{\text{MD}} = \frac{\bar{h}^{\text{fig}} - \bar{h}^{\text{lit}}}{\|\bar{h}^{\text{fig}} - \bar{h}^{\text{lit}}\|}. \quad (1)$$

By convention, positive scaling along v_{MD} (the steering vector) pushes activations toward the figurative class, negative scaling toward the literal class.

Probe (PROBE). Following Tigges et al. (2024), who define the steering direction as the unit-normalized weight vector of a logistic regression classifier, we train a binary logistic classifier:

$$p(c = \text{fig} \mid h) = \sigma(w^\top h + b), \quad (2)$$

where $w \in R^d$ is the model weight vector, $b \in R$ is the bias, $w^\top h$ denotes the dot product, and σ is the sigmoid function. The classifier is trained on the labeled activation set $H^{\text{fig}} \cup H^{\text{lit}}$, with figurative activations labeled 1 and literal activations labeled 0.

The steering vector is the unit-normalized weight vector: $v_{\text{PROBE}} = w/\|w\|$.

Like v_{MD} , v_{PROBE} orients toward the figurative class by construction. Probe-derived directions in over-parameterized regimes are known to be non-identifiable (Gao et al., 2026; Venkatesh and Kurupath, 2026); Appendix D reports per-model classification accuracy and provides additional analysis of the probe-derived steering directions.

Random (RAND). As a control for the possibility that any sufficiently large activation perturbation could shift the figurative-literal balance regardless of direction, we additionally evaluate $K_{\text{rand}} = 3$ random unit vectors, drawn uniformly from the unit hypersphere. Each random vector is applied under the same operator and scaling protocols as the learned vectors. RAND establishes the effect of generic activation perturbation independent of any learned direction.

4.2 Steering Operators

Given a steering vector v and an activation $h \in R^d$ at the target position, an operator returns a modified activation $h' = \mathcal{O}(h, v; \cdot)$. We evaluate two operators.

Additive ($\mathcal{O}_+$). Following Turner et al. (2024):

$$\mathcal{O}_+(h, v; \alpha) = h + \alpha v, \quad \alpha \in R. \quad (3)$$

Negative α pushes h along $-v$ (toward literal); positive α along $+v$ (toward figurative in our case). The additive operator changes both the direction and the norm of h .

Angular ($\mathcal{O}_\theta$). We also evaluate a norm-preserving angular operator inspired by Vu and Nguyen (2025). Instead of adding a scaled vector, angular steering rotates h within the input-dependent plane $\text{span}\{h, v\}$ by an angle $\theta \in [-\pi, \pi]$, where h is the activation at the intervention position and v is the unit-norm steering vector. The rotation controls intervention strength while preserving $\|h\|$. Since v is oriented toward the figurative class, negative θ rotates h toward v (figurative); positive θ rotates away (literal). Full details are given in Appendix E.

4.3 Application Protocol

We apply the steering operator at one input-token position p and one layer L_i , which we call the *injection layer*. The operator is implemented via a forward hook on the residual-stream output of layer L_i .

Token position. We apply the steering operator

at a single token position p : the final token of the idiomatic expression in the input sentence. This choice treats the idiom as a completed span, since under causal attention, the final token can incorporate information from the preceding context and earlier idiom tokens. We use this fixed position in all experiments; Appendix F discusses the rationale and alternative intervention sites considered during development.

Cross-layer construction. A central design choice is the separation of the *source layer* L_s , at which the steering vector is constructed (§4.1), from the *injection layer* L_i , at which the operator fires. We call $L_s = L_i$ *same-layer* steering (the standard choice) and $L_s \neq L_i$ *cross-layer* steering. The motivation is that representation quality and intervention leverage need not peak at the same layer: a direction may be easiest to estimate where the distinction is linearly accessible, yet most effective when injected at earlier layers that can still shape the continuation. We ablate L_s and L_i in §6.3.

KV-cache semantics. The intervention fires only once, on the prompt’s forward pass at position p : the modified residual stream at L_i propagates upward, producing modified key-value caches that subsequent generated tokens attend to in the standard way. This single-position, prompt-pass-only protocol differs from CAA (Rimsky et al., 2024), which intervenes at every post-prompt position throughout generation; it is closer in spirit to single-position causal interventions used in mechanistic interpretability. Appendix G gives the implementation details.

Generation settings. Each sentence prefix is passed verbatim to the model with no instruction or chat template. Completions are sampled at temperature 0.7, top- p 0.9, up to 50 new tokens, stopping at end-of-sequence or the token budget.

5 Experimental Settings

We evaluate five *base* LLMs from two architectural families, 2B–12B parameters: Llama-3.2-3B, Llama-3.1-8B, Gemma-2-2B, Gemma-3-4B, Gemma-3-12B. We use base models (prior to supervised fine-tuning) to isolate next-token generation behavior from instruction-following biases; instruction-tuned models are left to future work.

Steering grid. We evaluate two operators (additive, angular) crossed with two learned vector constructions (mean-difference, probe), plus three random unit-norm vectors as perturbation controls.

A *condition* is one (vector, operator, coefficient) combination, or the unsteered baseline. The intervention is global and input-independent. For additive steering, we parameterize intervention strength as $\alpha = \alpha_{\text{factor}} \cdot s_{\text{MD}}$, where α is the raw additive coefficient, α_{factor} is a model-normalized multiplier, and $s_{\text{MD}} = \|\bar{h}^{\text{fig}} - \bar{h}^{\text{lit}}\|$ is the norm of the raw mean-difference vector (with $\bar{h}^{\text{fig}}$ and $\bar{h}^{\text{lit}}$ the empirical class-mean activations). For each model, calibration on IdioLink selects the source layer L_s (where the steering vector is constructed) and the α_{factor} grid. IdiomSteer is not used during calibration. The injection layer L_i (where the operator fires) is fixed to $L_i = 2$. Angular steering is norm-preserving, so we use a shared grid of rotation angles $\theta \in \{\pm 60^\circ, \pm 70^\circ, \pm 90^\circ, \pm 120^\circ\}$ across all models. We sample 5 completions per sentence, giving 500 completions per condition. Full calibration details are in Appendix H.

Evaluation. Each generation is labeled *figurative*, *literal*, or *incoherent*. We report *literal rate* (proportion of generations labeled literal; the directionally harder task since figurative is the conventional default) and *coherence rate* (proportion of generations that are labeled either figurative or literal), with 95% bootstrap confidence intervals (CIs) over evaluation idioms (Appendix I). We scale evaluation with an LLM-as-a-judge (Zheng et al., 2023), calibrated against 280 Llama-3.2-3B generations labeled by two annotators (Cohen’s $\kappa = 0.867$). The selected judge, **Gemma-4-31B-it** (Google DeepMind, 2026), achieves 90.0% accuracy and $\kappa = 0.821$ against gold labels. Annotation guidelines, candidate judge models, and prompt iteration are in Appendix J.

6 Results and Discussion

6.1 RQ1: Activation Steering Shifts Figurative-vs-Literal Interpretation Across LLMs

We evaluate whether residual-stream activation steering can reliably shift LLM continuations toward literal interpretation of idioms. For each of five models—Llama-3.2-3B, Llama-3.1-8B, Gemma-2-2B, Gemma-3-4B, Gemma-3-12B—we summarize each method by the highest *literal rate* it attains on IdiomSteer within the fixed coefficient grid selected during calibration. This best-over-grid statistic measures steering capacity within a pre-specified evaluation grid; full curves over the grid are reported in Appendix K. Figure 2 com-

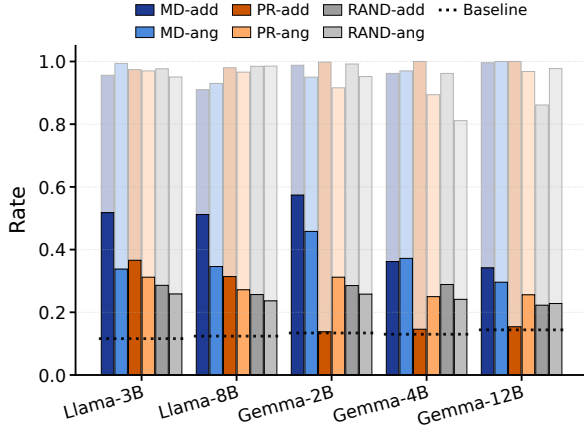


Figure 2: Maximum literal rate per (model, method) at the best operating point. Bars are stacked: solid (darker) = literal rate, lighter top = figurative-coherent rate; total height = coherence rate. Dotted line: unsteered baseline literal rate per model. Random results averaged over three unit-norm vectors. PR=PROBE; add=Additive; ang=Angular.

compares these best-over-grid operating points for each vector construction and steering operator.

Steering shifts the figurative-vs-literal balance across all five models. The unsteered baseline literal rate of 0.12–0.14 across models is itself evidence of the strong figurative default in idiom interpretation: even without intervention, the models overwhelmingly produce figurative continuations. Every learned method—mean-difference and probe, under both additive and angular operators—produces a literal-rate increase over the unsteered baseline on every model, though the magnitude varies substantially across constructions and families. Mean-difference additive steering is the dominant construction in four of five models, achieving a literal rate of 0.34–0.57 (a 2.4–4.5 $\times$ increase over the unsteered baseline); on Gemma-3-4B, the angular operator is marginally preferred (0.37 vs 0.36). Bootstrap confidence intervals for the headline operating points are reported in Appendix I. Probe-derived vectors exhibit a family-dependent pattern: PROBE-additive produces moderate effects on the Llama family (0.31–0.37) but collapses to near-baseline on the Gemma family (0.14–0.15 across all three Gemma models), consistent with the probe non-identifiability concern (§4.1, PROBE). PROBE-angular is more robust across families (0.25–0.31 on every model), suggesting that the rotation operator partially compensates for the probe vector’s weaker alignment with the figurative-vs-literal axis. Random unit-norm vectors produce shifts of 0.22–0.29 under both oper-

ators, smaller than the strongest learned method on every model. The angular operator does not improve random-vector performance over additive; we characterize the difference between directional and perturbative effects in Section 6.2.

The multiplier convention generalizes across norms. Across the five models, the norm of the raw (pre-normalization) mean-difference vector $\|\bar{h}^{\text{fig}} - \bar{h}^{\text{lit}}\|$ varies by three orders of magnitude (from 4.18 on Llama-3.2-3B to 7,134 on Gemma-3-12B). Despite this, the calibrated multiplier grids $\alpha_{\text{factor}} = \alpha/s_{\text{MD}}$ cluster within a factor of ~ 2.5 across all models, with the optimal operating point falling in $\alpha_{\text{factor}} \in [-2.4, -0.5]$ on every model (see Appendix H for more details). This suggests that the multiplier convention identifies a useful per-norm scale of intervention.

Summary (RQ1). Activation steering reliably shifts continuations toward literal interpretation across two model families with sizes ranging from 2B to 12B. Mean-difference additive steering is the most consistent construction, achieving roughly 2.4–4.5 $\times$ over baseline on every model. Optimal intervention strength is consistent across models in multiplier-normalized units; exceeding it trades literal-rate gains for fluency loss (Turner et al., 2024; Rimsky et al., 2024). Coherence at reported operating points remains ≥ 0.91 , indicating that literal-rate gains are achieved without sacrificing generation quality.

6.2 RQ2: The Steering Effect is Directional, not Perturbative

Whether the literal-rate shift reflects a genuine directional intervention or simply the destabilizing effect of a large activation perturbation is addressed with two diagnostic comparisons.

Sign-asymmetric literal rate. Figure 3 overlays the literal rate of mean-difference additive steering as a function of α_{factor} for all five models, with the random-vector average shown as a dashed control line. Mean-difference curves are strongly asymmetric: pushing in the literal direction (negative α_{factor}) produces large literal-rate shifts on every model, while pushing in the figurative direction leaves the literal rate at or below baseline. At the most extreme magnitudes tested, the negative-side literal rate is 3–4 $\times$ the positive-side rate on four of the five models, reflecting the model’s strong figurative default: there is room to shift toward literal but no room to shift further toward figurative. Random unit-norm vectors at matched magnitudes

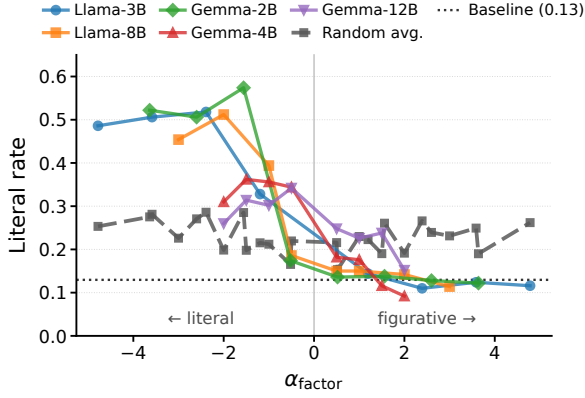


Figure 3: Literal rate as a function of α_{factor} for mean-difference steering, overlaid across all five models. Each curve is a separate model; the dashed grey line is the random-vector average. The unsteered baseline is shown as a dotted horizontal line. Strong asymmetry in α_{factor} sign is consistent across all five models, while random vectors show only mild magnitude-dependent shifts.

produce near-symmetric literal-rate shifts (0.18–0.29 across the five models on either sign) with no consistent direction, lacking the directional structure of mean-difference steering. This asymmetric mean-difference vs near-symmetric random pattern is visible across all five models in the per-model literal-rate curves in Appendix K.

Perplexity dissociation. A directional intervention should affect generation quality differently from a perturbative one. We measure per-token perplexity of steered generations under the unsteered model (Appendix L). On Llama-3.2-3B, the perplexity ratio $\text{PPL}_{\text{steered}}/\text{PPL}_{\text{baseline}}$ is roughly symmetric in α_{factor} sign (1.27 at -4.78 , 1.14 at $+4.78$), while the literal-rate effect is strongly asymmetric. Perplexity thus tracks intervention *magnitude*; literal rate tracks its *direction*. Random vectors show the same perplexity profile without producing comparable literal-rate asymmetry (Appendix L).

Summary (RQ2). The mean-difference steering effect is directional, not perturbative. Sign-asymmetric literal rate combined with sign-symmetric perplexity argues against a purely perturbative explanation in which any sufficiently large activation modification produces equivalent results; the asymmetry is consistent across all five models.

6.3 RQ3: Cross-layer Steering Substantially Outperforms Same-layer Steering

The application protocol of Section 4.3 separates the source layer L_s (where the steering vector is

constructed) from the injection layer L_i (where it is injected), with $L_s \neq L_i$ in our canonical setup. We ablate this choice with three layer-pair sweeps on three models spanning two families: Llama-3.2-3B, Llama-3.1-8B, and Gemma-2-2B. For each model we perform a source-layer sweep (varying L_s at fixed $L_i = 2$), an injection-layer sweep (varying L_i at the model’s canonical L_s), and a same-layer diagonal sweep ($L_s = L_i$). All sweeps hold α_{factor} fixed across source layers so that the steering-vector norm is matched within each sweep. Figure 4 shows the results; per-model heatmaps are in Appendix M.

Cross-layer steering substantially outperforms same-layer steering. At the matched coefficient shown in Figure 4 ($\alpha_{\text{factor}} = -2$), the cross-layer regime—mid-to-late source layer paired with $L_i \leq 2$ —achieves the highest literal rates across all three swept models: 0.55 on Llama-3.2-3B, 0.49 on Llama-3.1-8B, and 0.58 on Gemma-2-2B. Same-layer configurations reach only ~ 0.17 on each model, giving a $2.9\text{--}3.5\times$ gap. The optimal injection layer is $L_i \in \{0, 2\}$ and the optimal source layer falls in the mid-to-late range (50–75% of network depth) across all models.

Source-sweep shape and direction stabilization. Literal rate rises sharply with source layer through mid-network, peaks, then declines at the deepest layers alongside coherence. This is not a norm artifact: cross-layer cosines between mean-difference vectors (Appendix N) show the steering direction converges in the mid-to-late range (pairwise cosines 0.67–0.87) but diverges at the deepest layers (cosines 0.48–0.56 with the canonical layer), indicating the vector points in a genuinely different direction rather than simply growing in norm.

Mechanistic interpretation. One possible interpretation is that mid-to-late residual states make the figurative-vs-literal distinction more linearly accessible, while early-layer injection is most effective as it can still shape subsequent computation. Cross-layer steering may therefore work by estimating the direction where the distinction is easiest to recover and injecting it where the intervention has maximal downstream influence. We treat this as a hypothesis about layer-wise organization, rather than direct evidence for a fixed processing sequence.

Summary (RQ3). Cross-layer steering with L_s at mid-to-late depth and $L_i \in \{0, 2\}$ outperforms the best same-layer configuration by roughly $3\times$ on three models from two families. The regime is consistent across architectures, and direction-

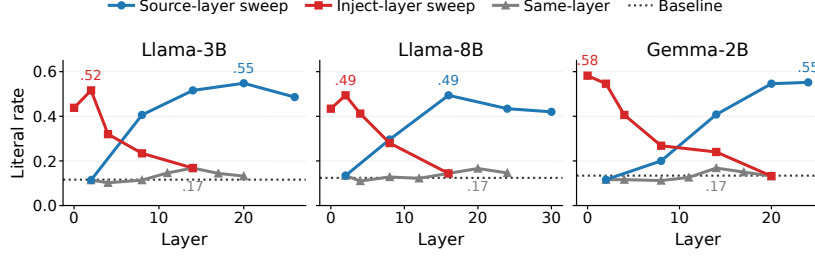


Figure 4: Layer-pair sweeps on three models at $\alpha_{\text{factor}} = -2$. Blue: source-layer sweep ($L_i = 2$ fixed); red: injection-layer sweep (canonical L_s fixed); grey: same-layer diagonal ($L_s = L_i$). Dotted line: unsteered baseline. Cross-layer steering outperforms same-layer steering by $2.9\text{--}3.5\times$.

stabilization in the mid-late residual stream, supported by mean-difference vector cosines, explains the location of the source-sweep optimum.

6.4 RQ4: The Steering Direction Captures an Idiom-General Property

The canonical setup already evaluates on idioms disjoint from training: steering vectors are computed on 22 IdioLink idioms and applied to the 20 disjoint idioms in IdiomSteer. We strengthen this with a cross-fold robustness analysis on Llama-3.2-3B. If the steering vector reflects a genuine property of how the model represents idiomaticity, re-deriving it from a different subset of idioms should produce a similar vector and similar steering behavior. We partition the 32 idioms from the IdioLink train and validation splits into three disjoint folds of 10–11 idioms each, using a shuffled round-robin assignment. For each fold, we re-derive mean-difference and probe steering vectors from that fold’s idioms alone and evaluate on IdiomSteer at the canonical layer configuration ($L_s = 14$, $L_i = 2$) with the additive operator.

Behavioral consistency. At $\alpha_{\text{factor}} = -4.78$ (the grid extreme, chosen to maximize steering signal in the cross-fold comparison), the three fold-derived vectors achieve literal rates of 0.48, 0.46, and 0.42, compared with 0.49 for the canonical 22-idiom vector. Thus, despite using roughly half the canonical idiom set, the folds vary by only 0.06 in literal rate. Full results are in Appendix O.

Vector-level agreement. Fold-derived mean-difference vectors have pairwise cosine $\{0.80, 0.71, 0.72\}$ (mean 0.74): different training subsets recover approximately the same direction in residual space. Probe-derived vectors do not converge across folds at the vector level (pairwise cosine ~ 0.28), but each fold’s probe vector retains substantial alignment (~ 0.59) with the mean-difference vector *computed on the same fold’s*

idioms, consistent with probe non-identifiability (Section 4.1, PROBE).

Summary (RQ4). The mean-difference vector is recoverable from any sufficiently diverse subset of idioms and produces equivalent behavioral effects on held-out evaluation idioms. The steering direction appears to capture an idiom-general property of the model’s residual stream, rather than a memorized aggregate of training-idiom features.

7 Conclusion

We have shown that the figurative-vs-literal interpretation of idiomatic expressions in LLMs is a property that admits inference-time control via residual-stream activation steering. Mean-difference steering reliably shifts continuations from the model’s default figurative reading toward literal interpretation across five base models from two architectural families, substantially increasing literal-rate over the unsteered baseline, with effects that are sign-asymmetric (ruling out generic activation perturbation), idiom-general (recoverable from disjoint training-idiom subsets), and consistent across model architectures when intervention strength is normalized by the raw mean-difference norm. The cross-layer protocol introduced here—constructing the vector at a mid-to-late source layer and injecting at the earliest layers—outperforms standard same-layer interventions by roughly $3\times$.

More broadly, our results highlight two diagnostics that may be useful for steering studies beyond idioms. First, separating the layer where a direction is estimated from the layer where it is injected can reveal effective cross-layer intervention regimes that same-layer steering would miss. Second, comparing sign-asymmetric behavioral effects against sign-symmetric perplexity changes helps distinguish semantically meaningful directions from generic activation perturbations.

Limitations

Our study, while providing evidence that residual-stream activation steering can control figurative-vs-literal idiom interpretation in LLMs, is subject to several limitations. First, we focus on English idioms and on base, pre-instruction-tuning language models. The choice of base models is intentional: because they are used directly as next-token predictors, their continuations provide a relatively direct view of how the model interprets an idiom-final prefix. This is central to the IdiomSteer design, where the model receives the sentence prefix verbatim, with no instruction, examples, or chat template, and the continuation reveals whether it follows the conventional figurative reading or the compositional literal reading. However, whether the same figurative-vs-literal direction can be recovered in other languages, particularly typologically distant ones where idioms may behave differently, and whether it persists after instruction tuning or RLHF, remain open questions.

Second, IdiomSteer is a controlled benchmark of 20 idioms with 5 sentence prefixes each, selected for bidirectional plausibility. This scale reflects the difficulty of the construction task: for each idiom, the prefix must end exactly at the idiomatic expression while leaving both the conventional figurative reading and the compositional literal reading naturally available. This constraint rules out many otherwise common idiom usages, where the preceding context already resolves the interpretation. The resulting benchmark allows us to isolate figurative-vs-literal continuation behavior under carefully validated ambiguity, but it does not exhaust the diversity of English idioms. Future work could extend the benchmark to a larger idiom inventory, additional syntactic forms, and broader contextual settings.

Third, our argument that the steering direction reflects a meaningful representation of figurative-vs-literal interpretation is supported by behavioral evidence—cross-architecture consistency, sign asymmetry, and generalization across held-out idioms—rather than by a complete causal account of how models naturally choose between figurative and literal readings. We do not test whether the same direction is necessary for ordinary, unsteered idiom interpretation outside the intervention regime.

Finally, the LLM-as-judge used to scale evaluation across thousands of generations was calibrated

on Llama-3.2-3B with mean-difference additive steering only, and then applied to all models and methods. This assumes that completion style does not vary across models or steering methods in ways that substantially affect judge decisions; stronger validation through per-model human annotation could test this assumption.

Ethics Statement

We use publicly available datasets and model checkpoints in accordance with their intended use and licenses, as detailed in Table 7 (Appendix P) and Table 8 (Appendix R). IdiomSteer consists of newly authored sentence prefixes and annotations created for this work. Although the idiom inventory was selected from MAGPIE (Haagsma et al., 2020), the benchmark examples themselves were manually written, curated, and validated by the authors. We will release IdiomSteer under a CC BY 4.0 license. Our use of AI assistants during writing and code and documentation support is disclosed in Appendix S.

No personally identifiable information is included in IdiomSteer, and annotators were instructed to flag and exclude examples containing personal, sensitive, or offensive content. Our work is intended solely for research purposes. Since we evaluate base language models, some generated continuations may contain offensive, biased, or otherwise inappropriate text. We mitigate this risk by using generations only for aggregate analysis and by not releasing raw model completions.

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A Qualitative Steering Examples

Table 1 shows qualitative examples of mean-difference steering in the literal direction: unsteered continuations typically follow the conventional idiomatic reading, while steered continuations use the idiom’s component words compositionally.

Continuation diversity. The examples in Table 1 illustrate a broader pattern: steered literal continuations are qualitatively diverse rather than formulaic. Across the full evaluation, literal completions vary in the compositional strategy they employ—some reinterpret the idiom’s head noun as a physical object (e.g., *ice*, *beans*), others extend a component word into an unexpected compound (*out of the blue room*; *take a bow and arrow*), and others construct novel scenarios around the literal meaning of the full phrase. We do not observe systematic mode collapse toward a single literal template for any idiom. A full quantitative analysis of continuation diversity, including type–token statistics and per-idiom breakdowns, will be released alongside the data repository.

B IdiomSteer Construction and Validation Details

Idiom selection. We manually selected 20 English idioms from MAGPIE (Haagsma et al., 2020). Selection was performed once, before any steering experiments, and was based only on whether the idiom could support five sentence prefixes satisfying our construction constraints. We did not select idioms based on anticipated steerability or on any property of the steering results. The IdiomSteer idioms are disjoint from all idioms used to estimate steering vectors.

Construction guidelines. Each sentence prefix is written to satisfy five constraints: (i) the target idiom appears at the end of the sentence, with no post-idiom context; (ii) the preceding context does not force either the figurative or literal interpretation; (iii) both figurative and literal continuations

are natural; (iv) the sentence is fluent and plausible under both readings; and (v) the five variants for each idiom differ in wording and scenario. The two readings are not required to be equally likely; the goal is bidirectional plausibility rather than balanced prior probability.

Sentence authoring. For each idiom, we manually authored five sentences, each ending in the idiomatic expression. We used Claude 3.5 Sonnet (Anthropic, 2024) only to propose candidate sentences during drafting. All final examples were curated, edited, and approved by the authors; no model-generated candidate was used without human revision and validation.

Validation protocol. Two independent annotators validated each candidate sentence using a dual-completion protocol. For each sentence, annotators were asked to: (i) produce a short natural continuation under a figurative interpretation; (ii) produce a short natural continuation under a literal interpretation; (iii) mark the sentence as *valid* if both continuations were natural, or *invalid* otherwise; and (iv) for invalid cases, propose a revision restoring bidirectional plausibility. Requiring annotators to produce both continuations is a stronger check than asking for an ambiguity judgment alone, since it forces each annotator to commit to a concrete reading before judging validity.

Iterative refinement. The benchmark is constructed and validated through several iterations. Sentences flagged as invalid by either annotator are revised—typically by changing the subject, or surrounding clause—and re-circulated. Iteration continued until all 100 sentences received the *valid* label from both annotators.

Idiom inventory. The 20 IdiomSteer idioms are: *up in the air*, *in hot water*, *out of the blue*, *break the ice*, *behind bars*, *on ice*, *see red*, *pull someone’s leg*, *at the crossroads*, *lie low*, *shut the door on*, *under the sun*, *take a bow*, *have a ball*, *in stitches*, *in the pink*, *green light*, *rough edges*, *in the fast lane*, *spill the beans*.

Final dataset. The final IdiomSteer benchmark contains 100 sentence prefixes, distributed evenly as five prefixes per idiom across 20 English idioms. All prefixes end at the target idiom and contain no post-idiom context. Each prefix was included only if both annotators judged that natural continuations were available under both the figurative and literal

Model	Unsteered continuation	Steered continuation
Gemma-2-2B	During the party, Sarah tried to <i>break the ice</i> by talking to the man next to her.	During the party, Sarah tried to <i>break the ice</i> with her bare hands, but the result was disastrous. The ice was so strong that she couldn't even break it. It was almost as if the ice had been glued together. Sarah was so frustrated that she didn't know what to do.
Llama-3.2-3B	This morning she called me <i>out of the blue</i> and said she had something to tell me.	This morning she called me <i>out of the blue</i> room and asked me to come to her room.
Llama-3.1-8B	He said nothing at first, but deep down he knew he was <i>in hot water</i> .	He said nothing at first, but deep down he knew he was <i>in hot water</i> and had to get out of there.
Gemma-3-4B	Before the competition started, he walked to the center to <i>take a bow</i> and greet the audience.	Before the competition started, he walked to the center to <i>take a bow</i> and arrow.
Gemma-3-12B	During the meeting he accidentally <i>spilled the beans</i> and said he was going to a secret location.	During the meeting he accidentally <i>spilled the beans</i> on the floor.

Table 1: Qualitative examples of mean-difference steering in the literal direction. Unsteered continuations typically follow the conventional idiomatic reading, while steered continuations use the idiom’s component words compositionally.

readings. The IdiomSteer idioms are disjoint from the idioms used to construct the steering vectors. The dataset, construction guidelines, and annotation guidelines will be released alongside the paper.

C Training Pool: IdioLink Subset

We construct steering vectors from the train split of IdioLink (Hashiloni et al., 2026), a benchmark containing figurative and literal uses of English idioms. We describe here only the subset used for steering-vector construction; the full dataset construction is described in the IdioLink paper.

Raw and filtered counts. The raw IdioLink train split contains 1,760 sentences across 22 idioms, uniformly distributed (30 figurative + 50 literal per idiom). Our filtering retains only sentences where the idiom string is literally present (no inflected forms), yielding 1,683 sentences (95.6% retention). The dropped 77 sentences mostly involve inflected or altered surface forms, with the largest counts for *see the light* (21), *go to town* (14), *out of pocket* (13), *get stuck in* (13), and *to boot* (7). Sentences in the retained pool average 107 characters, ranging from 67 to 180.

Label definitions. Each sentence is labeled as either *figurative*, where the idiom carries its conventional non-compositional meaning, or *literal*, where the component words are used compositionally outside the idiomatic frame. For example, a figurative usage may describe someone being *in hot water* with a committee, whereas a literal usage may describe a substance being dissolved in *hot water*.

Subject distribution. The raw training split is stratified across 10 subject categories: Politics, Sport, Technology, History, Medicine, Culture, Entertainment, Food, Business, and Environment, with 176 sentences per category before filtering.

Annotation provenance. The IdioLink train split combines human-annotated and weakly labeled examples: 271 of the 1,760 raw train sentences (15.4%) are gold-annotated, and the remaining 1,489 (84.6%) are weakly labeled.

Disjointness from evaluation. The 22 idioms used to construct the canonical steering vectors are disjoint from the 20 idioms in IdiomSteer. Therefore, all main evaluations are performed on idioms unseen during steering-vector construction.

The 22 training idioms. *cut and dried, get a grip, get stuck in, go to town, in black and white, in the can, in the driving seat, in your face, nuts and bolts, off the shelf, old hat, on paper, on the stump, on the wagon, out of pocket, out of the woods, over the top, point the finger, round the bend, see the light, to boot, under a cloud.*

D Probe construction and PCA analysis

This appendix expands on the brief note in Section 4.1.

Training details. The probe is a binary logistic regression $p(c = \text{fig} \mid h) = \sigma(w^\top h + b)$ fit on 1,683 training activations from IdioLink, with figurative activations labeled 1 and literal activations labeled 0. We use scikit-learn’s LogisticRegression with L2 regulariza-

tion ($C = 1.0$), the LBFGS solver, maximum 1,000 iterations, default tolerance (10^{-4}), and `fit_intercept=True` (the intercept is fit but not regularized by LBFGS). We do not standardize activations and do not rebalance classes (the 37%/63% imbalance is left as-is). The same hyperparameters are used across all five models; per-model convergence ranges from 30 to 322 iterations on the full fit (Table 2). No model triggered a non-convergence warning.

Training and held-out accuracy. Across all five models, the probe achieves 1.0 training accuracy at the canonical source layer. In the over-parameterized regime (d/n between 1.4 and 2.4), training accuracy is trivially achievable and uninformative on its own. We therefore additionally report 5-fold stratified cross-validation accuracy (Table 2). Held-out accuracy ranges from 0.969 to 0.975 across all five models, indicating that the figurative-vs-literal distinction is linearly decodable from residual-stream activations at L_s on every model. This consistency is notable in light of the headline finding in Section 6.1 that PROBE-additive collapses on the Gemma family: the probe *does* learn the distinction on Gemma activations, but the recovered weight vector does not function effectively as a steering direction. The disconnect between classification accuracy and steering effectiveness is the non-identifiability concern made concrete (Section 4.1; Venkatesh and Kurupath, 2026).

Model	d	L_s	Train acc.	5-fold CV	Iters
Llama-3.2-3B	3,072	14	1.000	0.975 ± 0.009	33
Llama-3.1-8B	4,096	16	1.000	0.975 ± 0.010	30
Gemma-2-2B	2,304	20	1.000	0.969 ± 0.007	86
Gemma-3-4B	2,560	17	1.000	0.969 ± 0.007	322
Gemma-3-12B	3,840	24	1.000	0.970 ± 0.006	220

Table 2: Per-model probe training accuracy and held-out 5-fold stratified CV accuracy. “Iters” is iterations to convergence on the full fit. All probes converged within the 1,000-iteration cap. Held-out accuracy is tight (0.969–0.975) across models, indicating that classification on held-out activations is consistent even though downstream steering effectiveness varies (Section 6.1).

Non-identifiability concern. The probe’s high held-out accuracy does not imply its weight vector w is unique. In over-parameterized regimes, many distinct w achieve comparable training accuracy, and the specific solution depends on initialization, regularization, and the activation set used for training (Gao et al., 2026; Venkatesh and Kurupath,

2026). This matters because we use w (not the probe’s predictions) as a steering direction.

PCA-based remediation. A natural fix is to reduce dimensionality before fitting, ensuring $n > d_{\text{reduced}}$. We compute the top- k principal components of the training activations $H^{\text{fig}} \cup H^{\text{lit}}$, project all activations onto the resulting k -dimensional basis, fit a logistic classifier with the same hyperparameters as the canonical probe on the projected representations, and reconstruct the steering direction in the original residual space by inverse-projecting the probe weight. We test $k \in \{50, 100, 500\}$ on Llama-3.2-3B.

Results. Table 3 reports the geometric properties of the resulting probe vectors. Three observations stand out. First, training accuracy degrades modestly as k decreases (1.000 at full dimension, 0.977 at $k = 50$), confirming that the full-dimensional probe is over-parameterized and that reducing dimensionality removes some optimization freedom. Second, the cosine between the PCA-reduced probe and the full-dimensional probe falls sharply with k (0.865 at $k = 500$, 0.611 at $k = 100$, 0.533 at $k = 50$): different dimensionality regimes recover materially different directions, consistent with probe non-identifiability. Third, and most informatively, the cosine between the PCA-reduced probe and the mean-difference vector *rises* as k decreases (0.484 at full dimension, 0.717 at $k = 50$). Reducing the dimensionality available to the probe pushes its solution toward the mean-difference direction.

Interpretation. The PCA results refine the non-identifiability picture. The full-dimensional probe is free to spread its decision boundary across many directions in residual space and finds a solution with cosine 0.484 to the mean-difference vector. Restricting the probe to fewer principal directions removes this freedom, and the recovered weight vector aligns more closely with the principal axis of class-conditional variation—i.e., the mean-difference direction. This is consistent with the cross-fold robustness results in Section 6.4: probe vectors trained on disjoint subsets of idioms have low pairwise cosine (~ 0.28) but each retains substantial alignment with its fold’s mean-difference vector (~ 0.59). The probe-derived direction is non-unique in high dimensions, but its instances share substantial projection onto the mean-difference axis, which is itself uniquely identified by the data.

Decision: keep the full-dimensional probe. We adopted the full-dimensional probe as our primary construction because the steering operator acts on the full residual stream regardless of the dimensionality used to estimate the steering direction; reconstructing a probe from a k -dimensional subspace risks discarding signal that lies in the residual basis. We do not report downstream steering effects for the PCA-reduced probes in this submission; the geometric analysis above is sufficient to characterize how dimensionality affects what the probe recovers.

k	var. expl.	acc.	$\ w\ $	cos w/ MD	cos w/ full
full	1.00	1.000	6.98	0.484	1.000
500	0.94	0.998	7.08	0.484	0.865
100	0.78	0.983	5.56	0.609	0.611
50	0.70	0.977	4.58	0.717	0.533

Table 3: Probe weight vector properties under PCA-based dimensionality reduction on Llama-3.2-3B at $L_s = 14$. “cos w/ MD” is the cosine with the mean-difference vector; “cos w/ full” is the cosine with the full-dimensional probe. Reduced-dimension probes diverge from the full-dim probe while converging toward the mean-difference direction.

E Angular Steering Operator

This appendix gives the full definition of the angular steering operator used in Section 4.2. Let d denote the residual-stream dimensionality, let $v \in R^d$ be a unit-norm steering vector, and let $h \in R^d$ be the residual-stream activation at the intervention position, i.e., the final idiom token at injection layer L_i . Following [Vu and Nguyen \(2025\)](#), the goal is to steer by rotating h rather than by adding a scaled vector. The intervention therefore changes the activation direction while preserving its norm.

We use an input-dependent rotation plane defined by the activation h and steering vector v . First, decompose h into its component parallel to v and its component orthogonal to v :

$$c_1 = v^\top h, \quad r = h - c_1 v, \quad c_2 = \|r\|. \quad (4)$$

Here, c_1 is the scalar coordinate of h along v , r is the residual component orthogonal to v , and c_2 is the norm of that orthogonal component. When $c_2 > 0$, define the second basis vector

$$b_2(h) = \frac{r}{\|r\|}. \quad (5)$$

Then $\{v, b_2(h)\}$ is an orthonormal basis for the two-dimensional plane $\text{span}\{v, h\}$, and h can be

written as

$$h = c_1 v + c_2 b_2(h). \quad (6)$$

The angular operator rotates this two-dimensional coordinate representation by an angle θ , measured in radians. Let c'_1 and c'_2 denote the rotated coordinates:

$$c'_1 = c_1 \cos \theta - c_2 \sin \theta, \quad (7)$$

$$c'_2 = c_1 \sin \theta + c_2 \cos \theta. \quad (8)$$

The modified activation is then

$$\mathcal{O}_\theta(h, v; \theta) = c'_1 v + c'_2 b_2(h). \quad (9)$$

In the degenerate case $c_2 = 0$, the activation is already parallel to v , so the plane $\text{span}\{v, h\}$ does not define a unique second basis direction. This case is negligible in practice for high-dimensional residual activations; in implementation, we treat it as a no-op and return h unchanged.

This construction has three useful properties. First, $\theta = 0$ gives the identity intervention. Second, for $c_2 > 0$, the operator is norm-preserving:

$$\|\mathcal{O}_\theta(h, v; \theta)\| = \|h\|. \quad (10)$$

Third, the intervention strength is controlled directly by the angular displacement $|\theta|$ within the rotation plane.

The sign of θ determines the rotation direction. Under the coordinate convention above, negative θ rotates h toward v , while positive θ rotates h away from v . Since our steering vector v is oriented toward the figurative class, negative θ pushes activations toward the figurative direction, and positive θ pushes them away from the figurative direction, which we treat as steering toward the literal side of the figurative-vs-literal axis. This sign convention is opposite to the additive operator, where negative α pushes along $-v$ and therefore toward the literal direction.

The choice between additive and angular steering is consequential. Additive steering is sensitive to the activation norm at the injection layer, which can vary substantially across layers and models ([Vu and Nguyen, 2025](#)). Angular steering instead decouples intervention strength from activation norm and bounds the intervention in the geometrically interpretable range $\theta \in [-\pi, \pi]$. We evaluate both operators to assess this trade-off empirically (Section 6).

F Token-Position Selection

In all reported experiments, steering is applied at a single token position p , corresponding to the final token of the idiomatic expression in the input sentence. For example, in “The leaders arrived early to *break the ice*,” the intervention is applied at the final token of *ice*.

This choice is motivated by the causal-attention structure of autoregressive transformers. At the final idiom token, the residual representation can integrate information from the preceding sentence context and from earlier tokens in the idiom. The final token therefore provides a natural intervention site for targeting the idiom as a completed semantic unit.

During development, we also considered other intervention sites, including the first idiom token and interventions over the full idiom span. First-token intervention is less directly aligned with the completed idiom, since later idiom tokens are not yet available at that position. Full-span intervention is also natural, but it introduces multiple perturbations per prompt and therefore changes the intervention from a single-site causal manipulation into a cumulative multi-site modification. To keep the intervention minimal and comparable across idioms of different token lengths, we use a single final-token intervention throughout.

We do not treat token-position choice as a controlled ablation in this paper. Instead, we fix the intervention site before the main experiments and focus our controlled analyses on vector construction, steering operator, layer choice, and idiom-level generalization.

G KV-cache implementation details

During autoregressive generation, the steering hook at injection layer L_i executes the following conditional logic at every forward pass:

- If the input sequence length exceeds the target position p (the position of the last prompt token), the hook overwrites the residual-stream activation h at position p with the steered activation.
- If the sequence length is 1 (i.e., only the most recently generated token is being processed, with prior tokens served from the KV cache), the condition $p < \text{seq_len}$ fails and the hook is a no-op.

The intervention therefore fires exactly once per generation: on the prompt’s first forward pass, at the canonical target position p . The modified residual stream at layer L_i propagates through subsequent layers $L_i+1, \dots, L_n$ during the same prompt pass, producing modified key-value caches at position p in every attention layer above L_i . All subsequent generated tokens then attend to these modified caches through the standard attention mechanism, without any further hook intervention.

This implementation differs from CAA (Rimsky et al., 2024), which applies the steering vector additively at every post-prompt token position throughout the generation. Our single-position, prompt-pass-only protocol is closer in spirit to the single-position interventions used in mechanistic interpretability, where the goal is to characterize the causal effect of an intervention at one residual-stream location rather than to maintain a sustained behavioral bias across many tokens.

H Experimental and Calibration Details

Motivation for multiplier scaling. The steering intervention in this paper is designed to be global and input-independent. For each model and source layer, we compute a single steering vector from the training pool, unit-normalize it, and reuse it unchanged for all evaluation examples. Likewise, the additive coefficient range is fixed per model: it is not recomputed as a function of the particular input sentence or the activation h being steered. This design is important for interpretability, since it tests whether a single direction and a fixed intervention scale can systematically control figurative-vs-literal interpretation.

However, different models have substantially different residual-stream magnitudes. A raw coefficient such as $\alpha = -20$ therefore does not correspond to the same relative intervention strength across models. To make coefficients comparable while preserving input-independence, we scale additive coefficients by the norm of the raw mean-difference vector computed at the source layer:

$$s_{\text{MD}} = \|\bar{h}^{\text{fig}} - \bar{h}^{\text{lit}}\|. \quad (11)$$

The final steering vector is unit-normalized, but s_{MD} is retained as a model-level scale parameter. Additive steering is then written as

$$\alpha = \alpha_{\text{factor}} \cdot s_{\text{MD}}. \quad (12)$$

Thus, α_{factor} denotes a model-normalized multiplier, while α denotes the raw additive coefficient applied in residual space.

Calibration procedure. For each model, we perform a short calibration procedure before the main evaluation, using only IdioLink training-pool sentences and never the IdiomSteer evaluation benchmark. Calibration selects two model-level hyperparameters: the source layer L_s used to compute the steering vector, and the fixed range of α_{factor} values used in the additive grid. The selected values are then frozen and applied to all IdiomSteer inputs.

The source-layer search is intentionally coarse rather than exhaustive. We evaluate a small set of candidate layers, with emphasis on middle layers, and choose a layer that produces a clear steering effect without severe fluency degradation on the calibration examples. In most models, the selected layer is approximately halfway through the network; Gemma-2-2B is the only exception, where the selected source layer is later in the network ($L_s = 20$, approximately 75% depth).

After selecting L_s , we sweep over additive multipliers and identify a range where steering produces a measurable behavioral effect while preserving usable generation quality. We then define a symmetric grid of eight α_{factor} values per model: four negative and four positive. The symmetric grid supports the directional-asymmetry analysis in Section 6.2.

The injection layer L_i is not selected during calibration. We fix $L_i = 2$ for all models in the main experiments, which keeps the application protocol constant across architectures and makes cross-model comparisons easier to interpret. The broader sensitivity to layer choice, including source-layer and injection-layer sweeps as well as same-layer versus cross-layer steering, is analyzed separately in Section 6.3.

Per-model configurations. The calibrated multiplier grids turn out to be comparable across models despite the three-orders-of-magnitude spread in s_{MD} . Table 4 reports the calibrated source layer, fixed injection layer, raw mean-difference norm, and additive coefficient grid for each model.

Across the five models, the union of all grids spans $\alpha_{\text{factor}} \in [-4.78, +4.78]$. Despite s_{MD} varying by three orders of magnitude, the calibrated multiplier ranges are within a factor of ~ 2.5 of one another. We take this cross-model consistency as

evidence that the multiplier convention identifies a meaningful per-norm scale at which steering is effective, abstracting over architectural differences in residual-stream magnitude.

Random-vector controls. Random-vector controls use the same parameter grids as the learned methods: for additive steering, each of the three random unit vectors is evaluated across all eight α_{factor} values in the model’s calibrated grid; for angular steering, each is evaluated across all eight θ values in the shared angle grid $\{\pm 60^\circ, \pm 70^\circ, \pm 90^\circ, \pm 120^\circ\}$. This yields $3 \times 2 \times 8 = 48$ random-control conditions per model.

Evaluation grid. The full evaluation grid contains 81 conditions per model: one unsteered baseline, four learned configurations evaluated at eight coefficient values each, and 48 random-control conditions. Each condition is evaluated on the full IdiomSteer benchmark with 5 sampled completions per sentence, yielding 500 generations per condition.

I Bootstrap confidence intervals

We estimate uncertainty over evaluation idioms via a block bootstrap. Each bootstrap sample draws 20 idioms with replacement from the 20 idioms in IdiomSteer; each sampled idiom contributes all five sentence variants and all generated completions associated with those sentences. We use $B = 1,000$ resamples with random seed 42 and report percentile 95% confidence intervals (the 2.5 and 97.5 percentiles of the bootstrap distribution) for the headline mean-difference additive operating points and the unsteered baseline. Table 5 reports the resulting intervals for each model.

J Judge Prompt and Validation

Judgment task. Each generated continuation is assigned one of three labels: *figurative*, *literal*, or *incoherent*. A continuation is labeled *literal* only when the idiom’s component words are used compositionally outside the conventional idiomatic frame. A continuation is labeled *figurative* when the idiom is used in its conventional idiomatic sense, even if that sense involves a physical action. For example, a completion in which an actor physically bows after a performance is still treated as figurative for *take a bow*, because the action realizes the conventional idiomatic frame. A continuation is labeled *incoherent* if it is syntactically

Model	L_s	L_i	s_{MD}	α_{factor} grid
Llama-3.2-3B	14	2	4.18	$\pm 1.20, \pm 2.39, \pm 3.59, \pm 4.78$
Llama-3.1-8B	16	2	4.15	$\pm 0.5, \pm 1.0, \pm 2.0, \pm 3.0$
Gemma-2-2B	20	2	96.18	$\pm 0.52, \pm 1.56, \pm 2.60, \pm 3.64$
Gemma-3-4B	17	2	2,299.87	$\pm 0.5, \pm 1.0, \pm 1.5, \pm 2.0$
Gemma-3-12B	24	2	7,134.37	$\pm 0.5, \pm 1.0, \pm 1.5, \pm 2.0$

Table 4: Per-model calibration and evaluation settings. L_s is the source layer used to construct the steering vector; L_i is the injection layer used in the main experiments.

defective or fails to describe an imaginable situation.

Human annotation sample. To validate the LLM-as-a-judge, we constructed a 280-item human annotation set from Llama-3.2-3B generations. For each of the 20 IdiomSteer idioms, we randomly selected one of the five sentence variants. For each selected sentence, we sampled two completions from each of seven steering conditions: the unsteered baseline, mean-difference additive steering at raw $\alpha \in \{-20, -10, +10, +20\}$, and a random-vector control at raw $\alpha \in \{-20, +20\}$. This gives $20 \times 7 \times 2 = 280$ annotated items. Items were shuffled and assigned blinded IDs, so annotators saw only the sentence prefix and completion, not the steering condition.

Human annotation protocol. Annotators labeled each item in two steps. First, they applied a reading-independent coherence check: the continuation was marked incoherent only if it was syntactically defective or failed to describe an imaginable situation, treating the idiom’s words as ordinary words if necessary. Second, for coherent continuations, annotators compared the figurative and literal readings of the idiom in context and selected the better-supported interpretation. If neither interpretation was clearly preferred, annotators defaulted to *figurative*, reflecting the conventional idiomatic reading.

Annotator information. Two authors annotated and revised the IdiomSteer examples and also participated in the human evaluation used to validate the LLM-as-a-judge. One annotator is a PhD student and the other is a Master’s student. Both performed the annotation as part of their work on the paper and received no additional monetary compensation. During judge-validation annotation, annotators were blinded to the steering condition. Difficult cases were resolved through discussion among the authors. Because both annotators are authors of the paper, they were aware that their annotations

would be used for dataset validation, judge validation, aggregate analysis, and release of the curated benchmark.

Human agreement. Two annotators independently labeled the 280 items. Inter-annotator agreement reached Cohen’s $\kappa = 0.867$. Disagreements were resolved jointly to produce gold labels. The final gold set contains 158 figurative, 95 literal, and 27 incoherent continuations. Per-label one-vs-rest agreement was $\kappa = 0.899$ for *figurative*, $\kappa = 0.858$ for *literal*, and $\kappa = 0.802$ for *incoherent*. Disagreements were concentrated near the expected decision boundaries: literal-vs-incoherent cases, where the continuation extended the idiom’s words in an awkward but potentially interpretable way, and literal-vs-figurative cases, where both readings remained plausible.

Candidate judges and prompt iteration. We evaluated several candidate judge models from both open and proprietary providers, including Gemini 2.5 Flash, GPT-4o, GPT-4o-mini, Claude Sonnet 3.5, Claude Haiku, DeepSeek, Llama-3.3-70B-Instruct, Qwen3, and Gemma-4-31B-it. We paired these models with approximately 14 prompt variants, selecting the model-prompt pair with the highest agreement against the human gold labels. Prompt development progressed from a direct single-label classification prompt to a structured-output prompt that requires the judge to reason separately about coherence, the figurative reading, and the literal reading before assigning the final label.

Prompt-design failure modes. Two recurring failure modes guided prompt iteration. First, earlier prompts sometimes conflated literal physical actions with literal interpretation. This is incorrect when the physical action is part of the conventional idiomatic meaning, as in *take a bow*. The final prompt therefore explicitly distinguishes using an idiom *as an idiom* from using its component words compositionally. Second, earlier prompts some-

times treated unusual literal uses as incoherent. The final prompt addresses this by applying coherence independently of idiom interpretation: a surprising literal continuation can be coherent if it describes an imaginable situation.

Final judge structure. The selected judge prompt elicits structured intermediate fields rather than only a single label. The judge first states the conventional figurative meaning of the idiom and the literal compositional meaning of its component words. It then applies each reading to the generated text, identifying cues that support or contradict each interpretation. The final three-way label is derived from these fields: *incoherent* if the continuation fails the coherence check; *literal* if the literal reading is clearly and substantially better supported than the figurative reading; and *figurative* otherwise.

Judge performance. The selected judge, **Gemma-4-31B-it** (Google DeepMind, 2026), achieved 90.0% accuracy (252/280) and Cohen’s $\kappa = 0.821$ against the human gold labels, close to the human-human agreement of $\kappa = 0.867$. Per-label F1 scores were 0.93 for *figurative*, 0.86 for *literal*, and 0.88 for *incoherent*. Coherent-vs-incoherent agreement was 97.5%, with only 7 of 280 samples crossing the coherent/incoherent boundary. Conditional on both the human gold label and the judge labeling a sample coherent, figurative-vs-literal agreement was 91.5% (227/248, $\kappa = 0.817$).

Calibration scope. The judge was selected using the 280-item Llama-3.2-3B annotation set described above. It was then applied to all models, vector constructions, operators, and coefficient values in the main evaluation. This assumes that completion style does not vary across models or steering methods in ways that substantially affect judge decisions; we discuss this limitation in the Limitations section.

Prompt release. The full human annotation guidelines, annotation form, judge prompt, structured-output schema, and decoding settings will be released alongside the code and data repository.

For reproducibility, Appendix Q reports implementation details, random seeds, hardware, software versions, and computational cost.

K Full Coefficient Sweeps

The headline results in Section 6.1 summarize each method by its best operating point within the calibrated coefficient grid (Table 4, Appendix H). To show that these results are not driven by a single selected coefficient, we report the full coefficient sweeps for each model below.

For additive steering, Figures 5, 7, 9, 11, and 13 show literal rate and coherence rate as a function of α_{factor} for each of the five models. Mean-difference steering exhibits a clear directional effect across all models: literal rate rises sharply for negative α_{factor} (the literal direction in our sign convention), reaching its maximum within the calibrated grid, while positive α_{factor} remains near the unsteered baseline. Probe additive steering varies in strength across model families (consistent with the family-dependent pattern reported in Section 6.1), while random vectors produce smaller and less directionally structured shifts on every model. Coherence remains high across most moderate coefficients, with degradation only at the largest-magnitude interventions.

For angular steering, Figures 6, 8, 10, 12, and 14 show the corresponding sweep over rotation angles θ . Under our angular sign convention, positive θ rotates activations away from the figurative direction and corresponds to literal steering. Mean-difference angular steering increases literal rate over baseline across a range of positive angles on every model, while very large rotations reduce coherence. Probe and random angular steering show weaker gains and stronger sensitivity to rotation magnitude.

Together, these sweeps support the use of best-over-grid headline summaries in Section 6.1: the reported effects reflect broad regions of effective steering within the calibrated grid, rather than isolated outlier coefficients.

L Perplexity analysis

To complement the judge-derived coherence metric used in our main results, we compute per-token perplexity on the canonical Llama-3.2-3B run. For each generation, we score the steered output under the *unsteered* Llama-3.2-3B (i.e., we measure how surprising the steered generation is to the unmodified model). We report token micro-perplexity (per-token, robust to varying completion lengths) and the perplexity ratio $\text{PPL}_{\text{steered}}/\text{PPL}_{\text{baseline}}$, following the convention of Turner et al. (2024). We

Model	Baseline	MD-add literal	MD-add coh.
Llama-3.2-3B	0.116 [0.070, 0.170]	0.518 [0.442, 0.594]	0.956 [0.934, 0.976]
Llama-3.1-8B	0.124 [0.072, 0.182]	0.512 [0.440, 0.586]	0.910 [0.848, 0.958]
Gemma-2-2B	0.134 [0.088, 0.184]	0.574 [0.490, 0.662]	0.988 [0.974, 0.998]
Gemma-3-4B	0.130 [0.074, 0.188]	0.362 [0.280, 0.444]	0.962 [0.940, 0.980]
Gemma-3-12B	0.144 [0.086, 0.210]	0.342 [0.268, 0.418]	0.996 [0.990, 1.000]

Table 5: Bootstrap 95% confidence intervals (percentile) for the unsteered baseline literal rate and the headline mean-difference additive operating point per model. “Baseline” is the unsteered literal rate; “MD-add literal” and “MD-add coh.” are the mean-difference additive literal rate and coherence rate at the per-model best operating point.

aggregate over 500 completions per condition (100 sentences $\times$ 5 samples). No condition produces a degenerate (infinite-loss) generation.

Table 6 reports the literal rate, coherence rate, and perplexity ratio for the principal methods at four representative steering strengths on Llama-3.2-3B.

Method	α_{factor}	Lit.	Coh.	PPL ratio
baseline	—	0.12	1.00	1.00
mean-difference	−4.78	0.49	0.84	1.27
mean-difference	−2.39	0.52	0.95	1.10
mean-difference	+2.39	0.13	0.98	1.09
mean-difference	+4.78	0.13	0.97	1.14
probe	−2.39	0.37	0.95	1.06
probe	−4.78	0.34	0.79	1.15
random (avg.)	−4.78	0.25	0.86	1.17
random (avg.)	+4.78	0.26	0.91	1.10

Table 6: Behavioral effects and perplexity on Llama-3.2-3B at $L_s = 14$, $L_i = 2$. Literal rate (**Lit.**) and coherence rate (**Coh.**) are the judge-derived metrics defined in Section 5. Perplexity ratio (**PPL ratio**) is the token micro-perplexity of the steered generation under the unsteered model, normalized by the baseline perplexity (4.77). Random results are averaged over three independent unit-norm vectors.

Two observations from the table. First, perplexity rises modestly with steering: at the best literal point ($\alpha_{\text{factor}} = -2.39$ for Llama-3.2-3B), mean-difference achieves a perplexity ratio of 1.10, a 10% relative increase over baseline. The strongest steering tested ($\alpha_{\text{factor}} = -4.78$) reaches 1.27, still well within the range of fluent generation as confirmed by the judge-coherence rate (0.84 at this point).

Second, perplexity tracks intervention magnitude, not direction. At $\alpha_{\text{factor}} = -4.78$ vs $+4.78$, mean-difference perplexity ratios are 1.27 and 1.14—comparable in magnitude—while the literal rate is 0.49 vs 0.13, a $3.8\times$ asymmetry. Random

vectors at the same magnitudes produce a comparable perplexity pattern (1.17 vs 1.10) and near-symmetric literal rates (0.25 vs 0.26). This dissociation supports the Section 6.2 interpretation: mean-difference identifies a directional axis in residual space, while perplexity reflects only the perturbative magnitude of any sufficiently large activation modification.

M Layer-sweep heatmaps

Figure 15 shows the evaluated (L_s, L_i) grid for Llama-3.1-8B at four multiplier values. Dotted cells denote layer pairs that were not evaluated. The cross-layer advantage is visible as a hot region in the bottom rows ($L_i \leq 2$) at mid-to-late source layers; the diagonal is uniformly weak. The contrast between negative and positive multiplier panels mirrors the sign asymmetry reported in Section 6.2.

N Cross-layer steering vector cosines

The source-sweep curves in Section 6.3 show a characteristic rise-peak-decline shape: literal rate climbs through the mid-network, peaks at the canonical source layer, and falls at the deepest tested layers. A natural concern is whether the rise-peak-decline pattern reflects something about *where* the steering direction is best recovered, or whether it is an artifact of how the multiplier convention interacts with growing residual-stream norms.

To distinguish these, we compute pairwise cosine similarities between mean-difference steering vectors derived at different source layers. Two vectors trained on the same data at different source layers may point in similar directions (large positive cosine) if both layers encode the figurative-vs-literal distinction in approximately the same way, or in different directions (low cosine) if the layers carry distinct representations. Figure 16 shows the resulting pairwise cosine matrices for the three

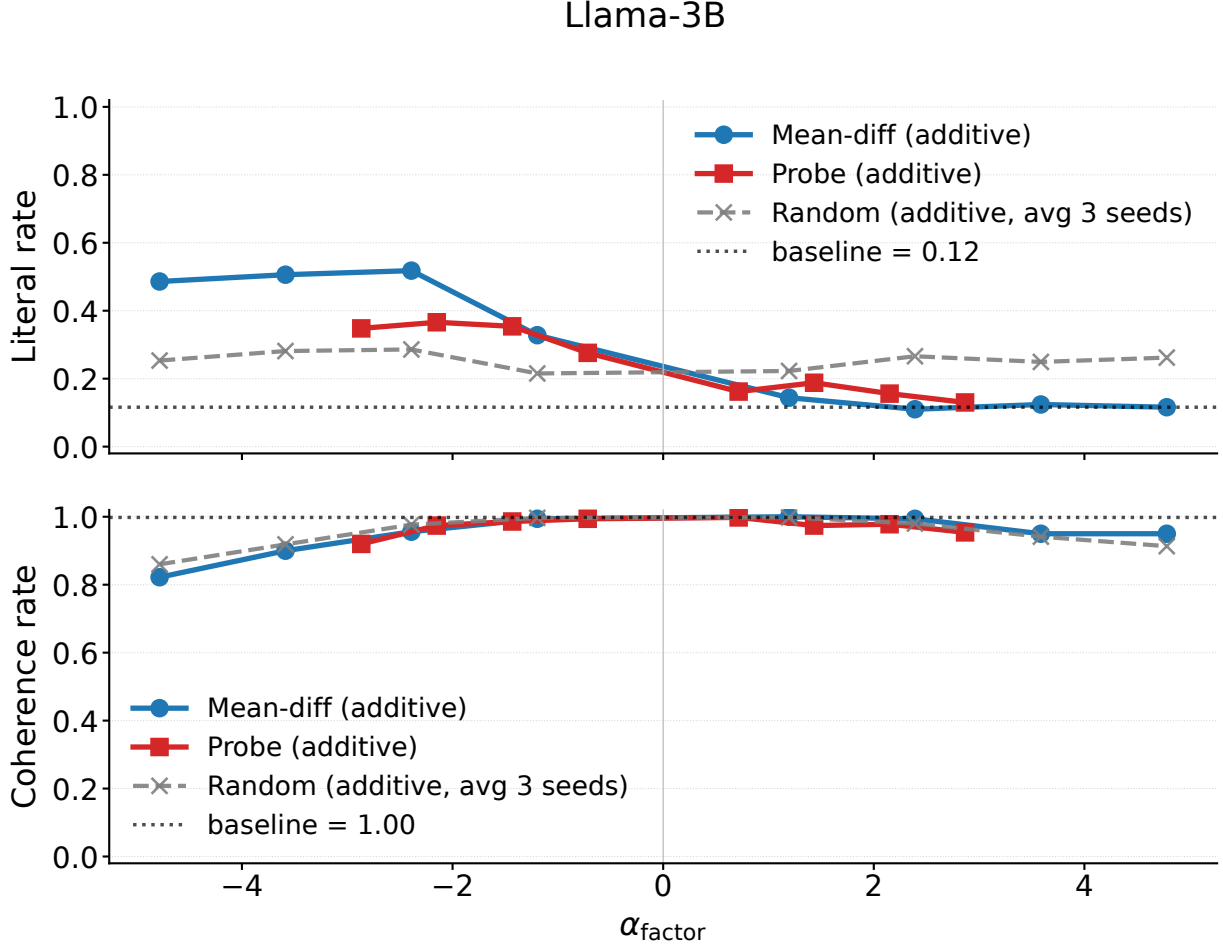


Figure 5: Full additive coefficient sweep on Llama-3.2-3B. Top: literal rate vs α_{factor} for mean-difference, probe, and random-vector steering. Bottom: coherence rate.

swept models.

The pattern is consistent across all three swept models: (i) the steering vector at the earliest layer ($L_s = 2$) is roughly orthogonal to vectors at all deeper layers, indicating that very early layers do not encode a recoverable figurative-vs-literal direction; (ii) the vectors at the canonical source layer and its nearest neighbors form a high-cosine cluster, with pairwise cosines of 0.67–0.87 (Llama-3.2-3B), 0.68–0.87 (Llama-3.1-8B), and 0.79–0.87 (Gemma-2-2B); (iii) at the deepest tested source layer, the vector drifts from the canonical cluster, most pronounced on the Llama models (cosines 0.48–0.56 with the canonical layer) and milder on Gemma-2-2B (cosine 0.79).

This third observation is the relevant one for interpreting the source-sweep decline. At the deepest source layers, the literal-rate drop is not merely a function of larger steering-vector norms (which we have already normalized through α_{factor}): the vector is also pointing in a measurably different

direction, with substantially weaker overlap with the canonical figurative-vs-literal axis. The rise-peak-decline shape therefore reflects a meaningful property of how the model represents figurative-vs-literal at different processing depths, not a residual norm confound.

O Cross-fold Robustness Curves

This appendix shows the literal-rate-vs- α_{factor} curves for the cross-fold robustness analysis (Section 6.4). Figure 17 overlays the curves for the canonical 22-idiom run (four grid points) and the three folds (each trained on 10–11 disjoint idioms and evaluated at $\alpha_{\text{factor}} \in \{\pm 3.59, \pm 4.78\}$). The fold curves track the canonical curve closely at the tested points, indicating that the steering effect is robust to the specific choice of training idioms.

P Model Checkpoints

Table 7 lists the base-model checkpoints used for the steering experiments. All five models are eval-

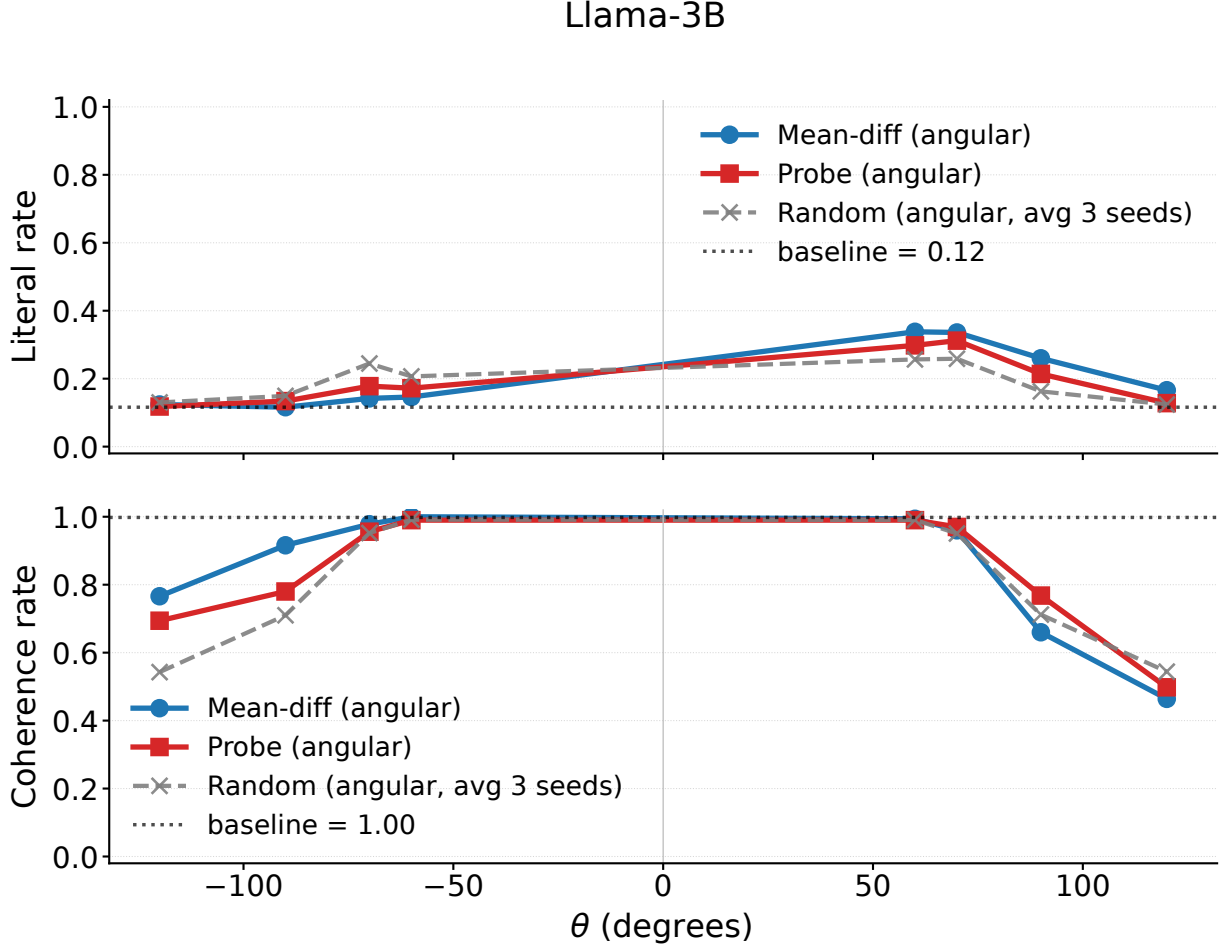


Figure 6: Full angular coefficient sweep on Llama-3.2-3B. Top: literal rate vs rotation angle θ . Bottom: coherence rate.

uated in their pre-instruction-tuning form.

Auxiliary models. Claude 3.5 Sonnet was used only to propose candidate IdiomSteer sentence prefixes during dataset drafting; all final examples were manually curated, edited, and validated by the authors. Gemma-4-31B-it was used as the LLM-as-a-judge after validation against human annotations (Appendix J).

Q Reproducibility and computational cost

This appendix documents implementation details, hyperparameter choices, random seeds, and computational cost for reproducibility.

Hardware and inference setup. All experiments run on a single host with four NVIDIA RTX 3090 GPUs (24 GB each). Llama-3.2-3B, Llama-3.1-8B, Gemma-2-2B, and Gemma-3-4B each fit on a single GPU; Gemma-3-12B is loaded across two GPUs via HuggingFace `device_map="auto"`. We

use HuggingFace transformers 4.55.0 with PyTorch 2.5.1 (CUDA 12.1 build). Llama-3.2-3B, Llama-3.1-8B, and Gemma-2-2B are loaded in fp16; Gemma-3-4B and Gemma-3-12B in bf16. Attention implementation defaults are kept for the Llama and Gemma-2 families; the Gemma-3 family uses `attn_implementation="sdpa"`. KV-cache is enabled at inference (HuggingFace default). Inference batch size is 1 (sentence-by-sentence forward-and-generate loop).

Layer indexing. We refer to layers via the HuggingFace `output_hidden_states` convention: `hidden_states[0]` is the embedding output, and `hidden_states[N]` is the output of physical decoder block $N-1$. The source layer L_s uses this index for activation collection. The injection hook attaches to `model.model.layers[L_i]` via a forward hook on the full block output (post-attention + post-MLP residual stream). $L_i = 2$ therefore modifies the output of decoder block 2.

Llama-8B

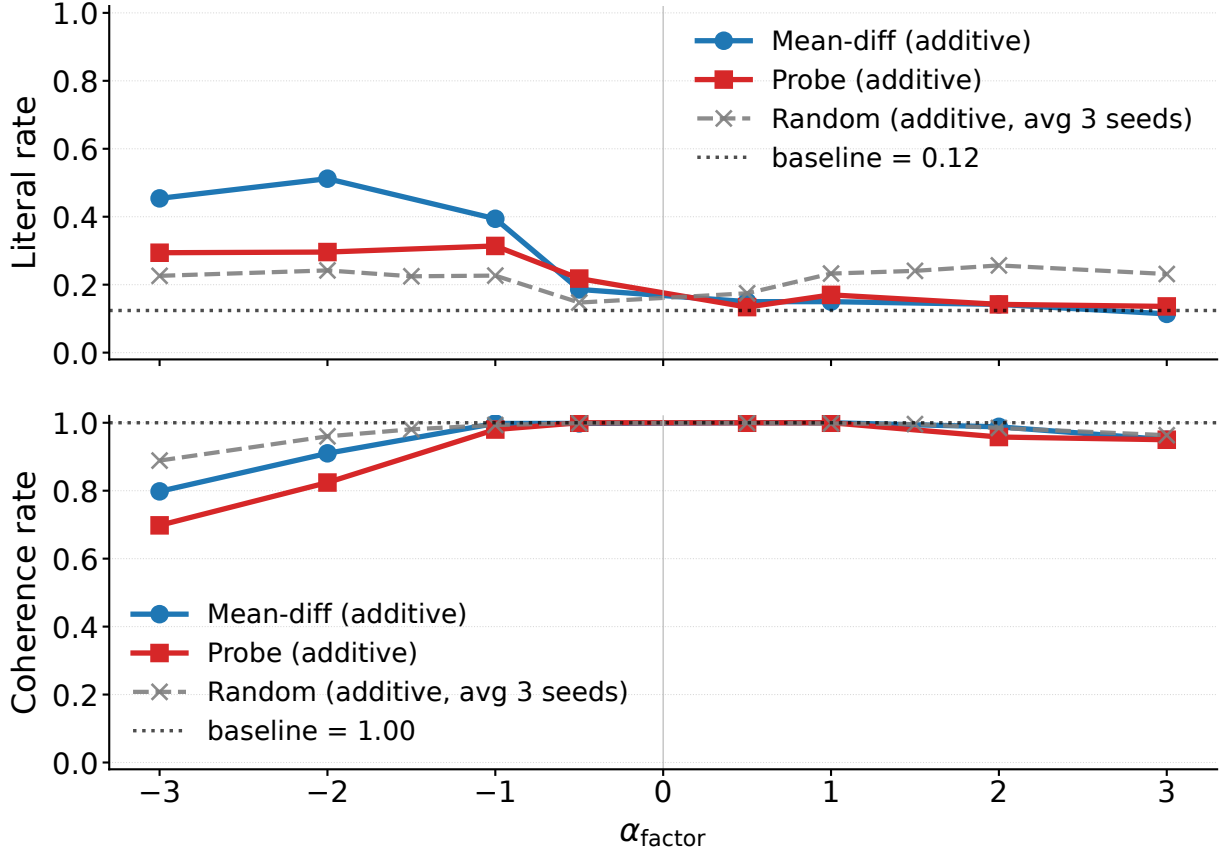


Figure 7: Full additive coefficient sweep on Llama-3.1-8B. Top: literal rate vs α_{factor} . Bottom: coherence rate.

Activation and steering vector precision. Activations are collected in the model’s native dtype (fp16 or bf16) and cast to fp32 before being stacked into the design matrix. The mean-difference vector v_{MD} and the probe weight vector are computed in fp32. At inference, the additive operator adds $\alpha \cdot v$ in the residual stream’s native dtype; the angular operator casts intermediate quantities through fp32 and back to fp16/bf16 before writing to the residual stream. Steering vectors are not persisted to disk; they are recomputed on every evaluation run from the same IdioLink training set, ensuring bit-for-bit reproducibility given the fixed seed.

Generation and random seeds. We set `torch.manual_seed(42)`, `numpy.random.seed(42)`, and `random.seed(42)` once at the start of each evaluation run; the seed is not re-set per condition. The 5 completions per sentence are produced by a sequential loop of 5 `generate()` calls drawing from the advancing RNG state (not `num_return_sequences=5`). This is deterministic across reruns given fixed

hardware and library state; we do not enable `torch.use_deterministic_algorithms(True)`, so cuBLAS/SDPA nondeterminism can introduce small variation across distinct hardware. The three random unit-norm control vectors (Section 4.1) are generated with `np.random.RandomState(s).randn(d)` for $s \in \{42, 123, 7\}$, yielding three different vectors per model (because d varies) under the same three seeds. The bootstrap procedure (Section 5, Appendix I) uses $B = 1000$ resamples with seed 42. The cross-fold shuffle (Section 6.4) uses seed 42; only a single shuffle was performed.

Probe hyperparameters. The probe uses scikit-learn 1.6.1 LogisticRegression with L2 regularization at $C = 1.0$ (the sklearn default), LBFGS solver, maximum 1,000 iterations, tolerance 10^{-4} , and `fit_intercept=True`. We did not perform a hyperparameter search over C , regularization type, or solver. Per-model classification accuracy is reported in Appendix D, Table 2.

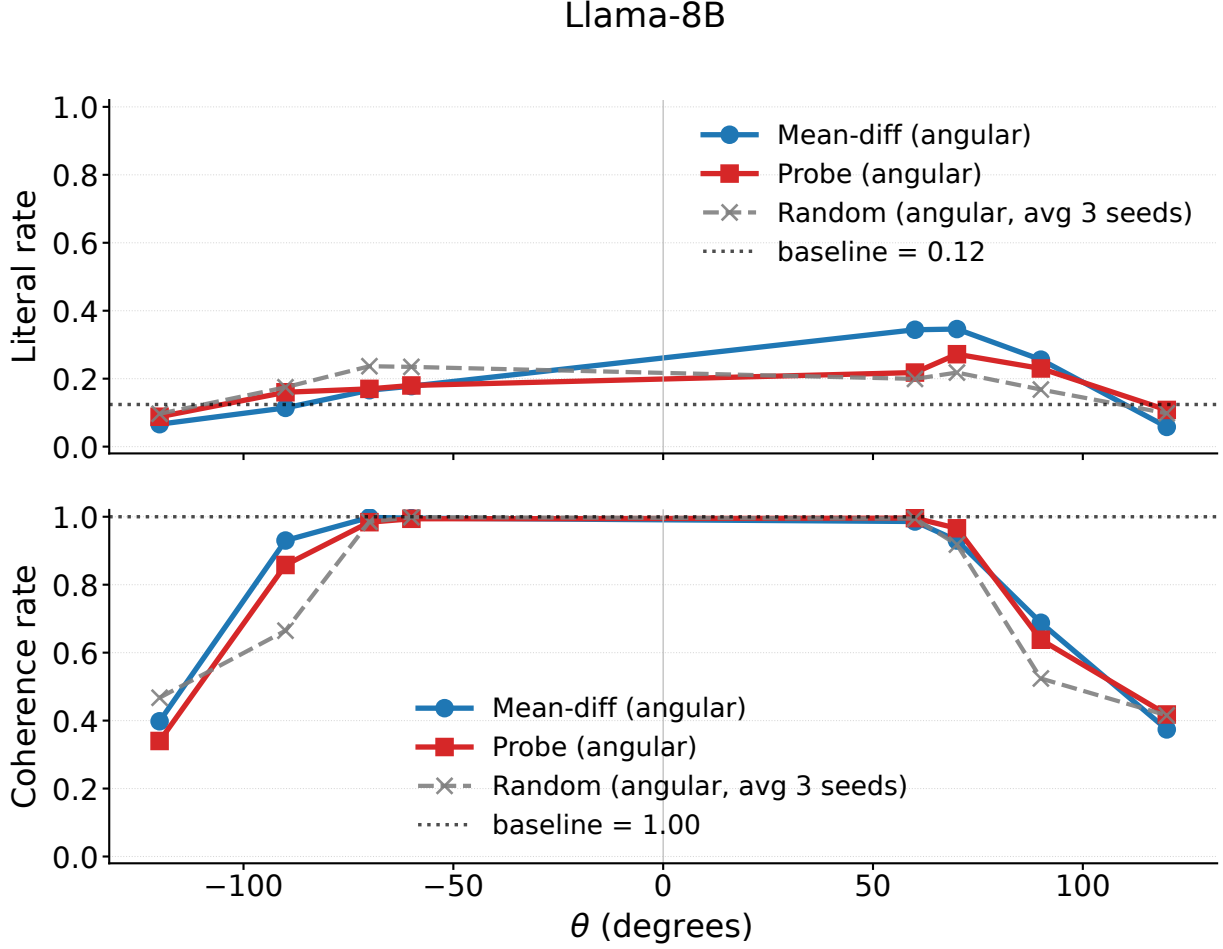


Figure 8: Full angular coefficient sweep on Llama-3.1-8B. Top: literal rate vs θ . Bottom: coherence rate.

Calibration. For each model, we manually select the source layer L_s by inspecting steered generations on a small subset of IdioLink training-pool idioms (three sentences per training idiom). Selection is qualitative: L_s is chosen to produce visibly literal completions without obvious fluency degradation. The α_{factor} grid is then expanded around the selected layer to span a literal-rate plateau and a fluency-loss regime, yielding the eight α_{factor} values per model reported in Appendix H. The IdiomSteer benchmark is never used during calibration.

Compute cost. The main evaluation grid (81 conditions $\times$ 500 completions per condition per model $\times$ 5 models = 202,500 generations) runs in approximately 85.7 GPU-hours: 5.96 h for Llama-3.2-3B, 9.24 h for Llama-3.1-8B, 9.27 h for Gemma-2-2B, 13.68 h for Gemma-3-4B, and 23.79 h $\times$ 2 GPUs for Gemma-3-12B. The cross-fold robustness evaluation (Section 6.4) adds 7.06 GPU-hours on Llama-3.2-3B. The layer-pair sweeps (Section 6.3) add approximately 13 GPU-hours across three models. Calibration runs across all

models add an estimated further 10–20 GPU-hours not separately tabulated. Total documented compute is approximately 106 GPU-hours, with up to 20 GPU-hours additional calibration overhead. The LLM-as-judge inference uses Gemma-4-31B-it via the OpenRouter API at temperature 0.0, `max_tokens = 700`; across all phases of the project, the judge was called approximately 358,700 times at a documented total cost of \$91.77.

Dataset splits. The IdioLink training pool contains 1,683 sentences across 22 idioms (629 figurative, 1,054 literal), drawn from the IdioLink train split after surface-form filtering (Appendix C). The cross-fold robustness analysis (Section 6.4) uses the union of the IdioLink train and validation idiom sets (32 idioms total) and partitions them into three disjoint folds of 11, 11, and 10 idioms. The IdiomSteer evaluation benchmark contains 100 sentences across 20 idioms (five variants per idiom). The IdiomSteer idiom set is disjoint from the IdioLink train and validation sets; this disjointness is asserted in code.

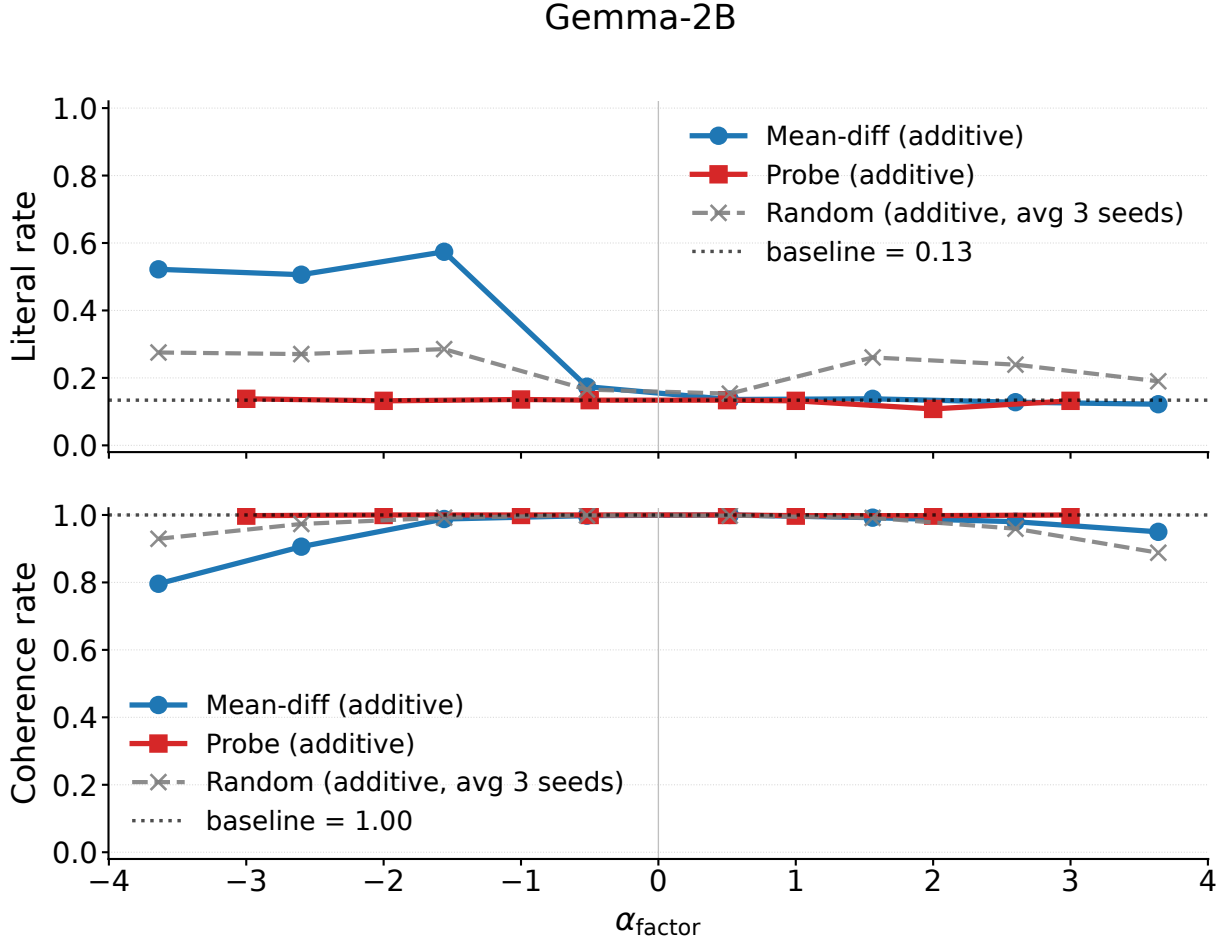


Figure 9: Full additive coefficient sweep on Gemma-2-2B. Top: literal rate vs α_{factor} . Bottom: coherence rate.

R Artifacts

Table 8 lists the datasets, software packages, and API providers used in this work. Model checkpoints are listed separately in Appendix P. Exact package versions are provided in the released repository.

We use Hugging Face Transformers and PyTorch for model loading, activation extraction, and generation; scikit-learn for probe training and PCA analysis; and OpenRouter for LLM-as-a-judge API access. IdiomSteer is newly created in this work and will be released under a CC BY 4.0 license. All external datasets, model checkpoints, and software artifacts are used for research purposes consistent with their licenses and terms of use.

S AI assistants

We used AI assistants (e.g., ChatGPT) to support code formatting, phrasing suggestions, and LaTeX styling during writing. Claude 3.5 Sonnet was used separately to propose candidate sentences during IdiomSteer drafting, as described in Section 3 and

Appendix B. All outputs were reviewed and edited by the authors. No content was directly generated or used without human verification.

Gemma-2B

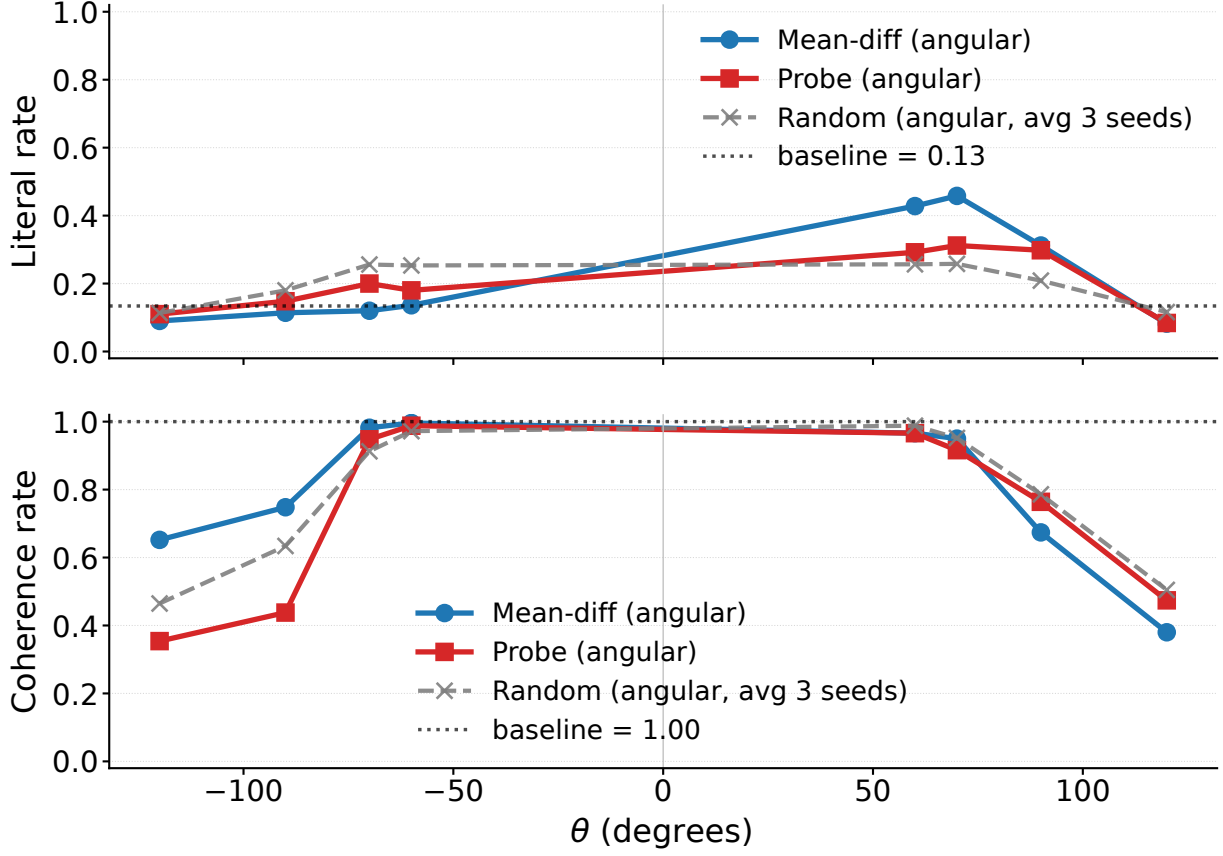


Figure 10: Full angular coefficient sweep on Gemma-2B. Top: literal rate vs θ . Bottom: coherence rate.

Model	Checkpoint	License	# Param.
Llama-3.2-3B	meta-llama/Llama-3.2-3B	Llama 3.2 Community License	3B
Llama-3.1-8B	meta-llama/Llama-3.1-8B	Llama 3.1 Community License	8B
Gemma-2-2B	google/gemma-2-2b	Gemma Terms of Use	2B
Gemma-3-4B	google/gemma-3-4b-pt	Gemma Terms of Use	4B
Gemma-3-12B	google/gemma-3-12b-pt	Gemma Terms of Use	12B

Checkpoints correspond to the public pretrained/base variants used for continuation generation and activation extraction. Instruction-tuned variants are not used in the main steering experiments.

Table 7: Base-model checkpoints used in the steering experiments.

Gemma-4B

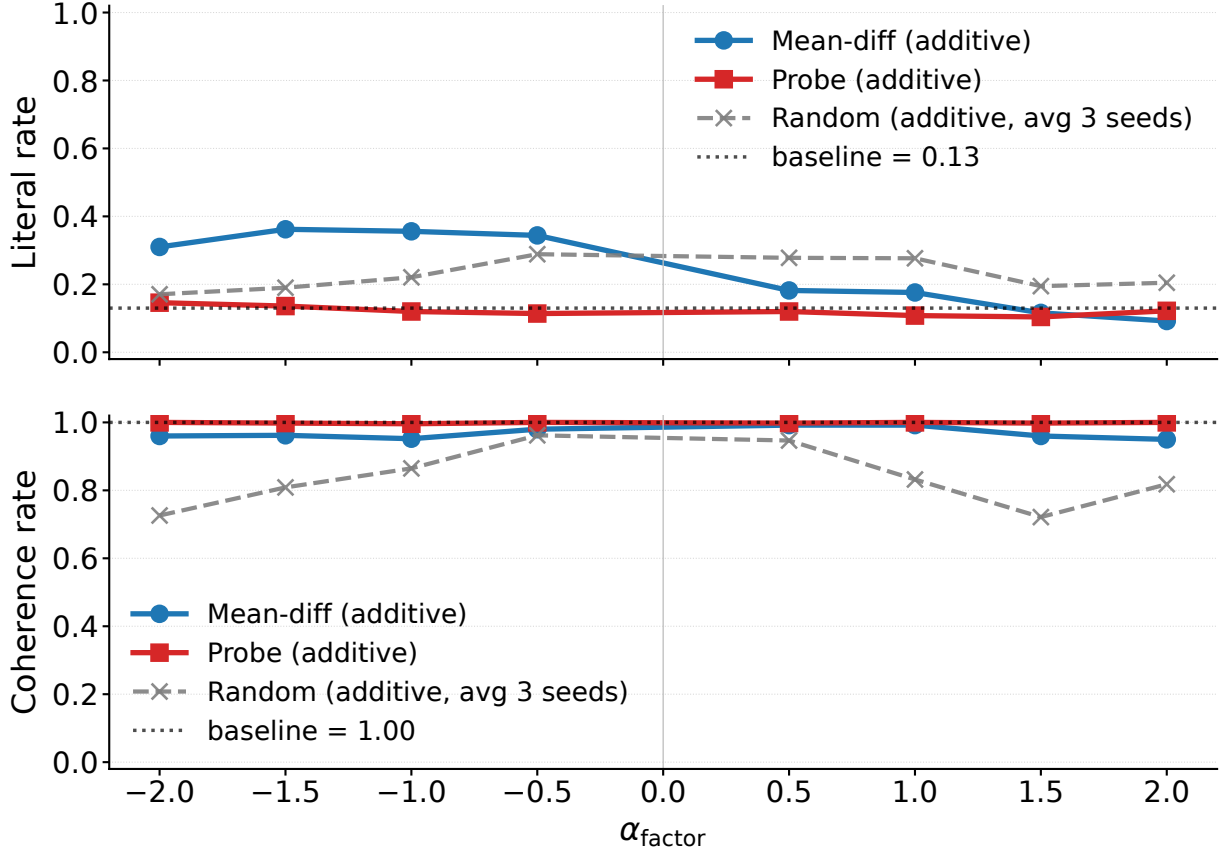


Figure 11: Full additive coefficient sweep on Gemma-3-4B. Top: literal rate vs α_{factor} . Bottom: coherence rate.

Artifact	Type	License / Terms	Usage
IdiomSteer	Dataset	CC BY 4.0	Evaluation benchmark introduced in this work.
MAGPIE ^a	Dataset	CC BY 4.0	Source idiom inventory for selecting IdiomSteer idioms.
IdioLink ^b	Dataset / Code repository	Apache-2.0	Training pool used to construct steering vectors.
HuggingFace Transformers ^c	Software library	Apache-2.0	Model loading, activation extraction, and generation.
PyTorch ^d	Software library	BSD-style license	Tensor computation, model execution, and forward hooks.
scikit-learn ^e	Software library	BSD 3-Clause	Logistic-regression probes, PCA analysis, and cross-validation.
OpenRouter ^f	API provider	Proprietary terms	API access for LLM-as-a-judge evaluation.

^a <https://github.com/hslh/magpie-corpus>

^b Repository URL withheld for anonymous review; will be provided in the camera-ready version.

^c <https://github.com/huggingface/transformers>

^d <https://github.com/pytorch/pytorch>

^e <https://github.com/scikit-learn/scikit-learn>

^f <https://openrouter.ai/terms>

Table 8: Datasets, software packages, and API providers used in this work. Model checkpoints are reported separately in Table 7.

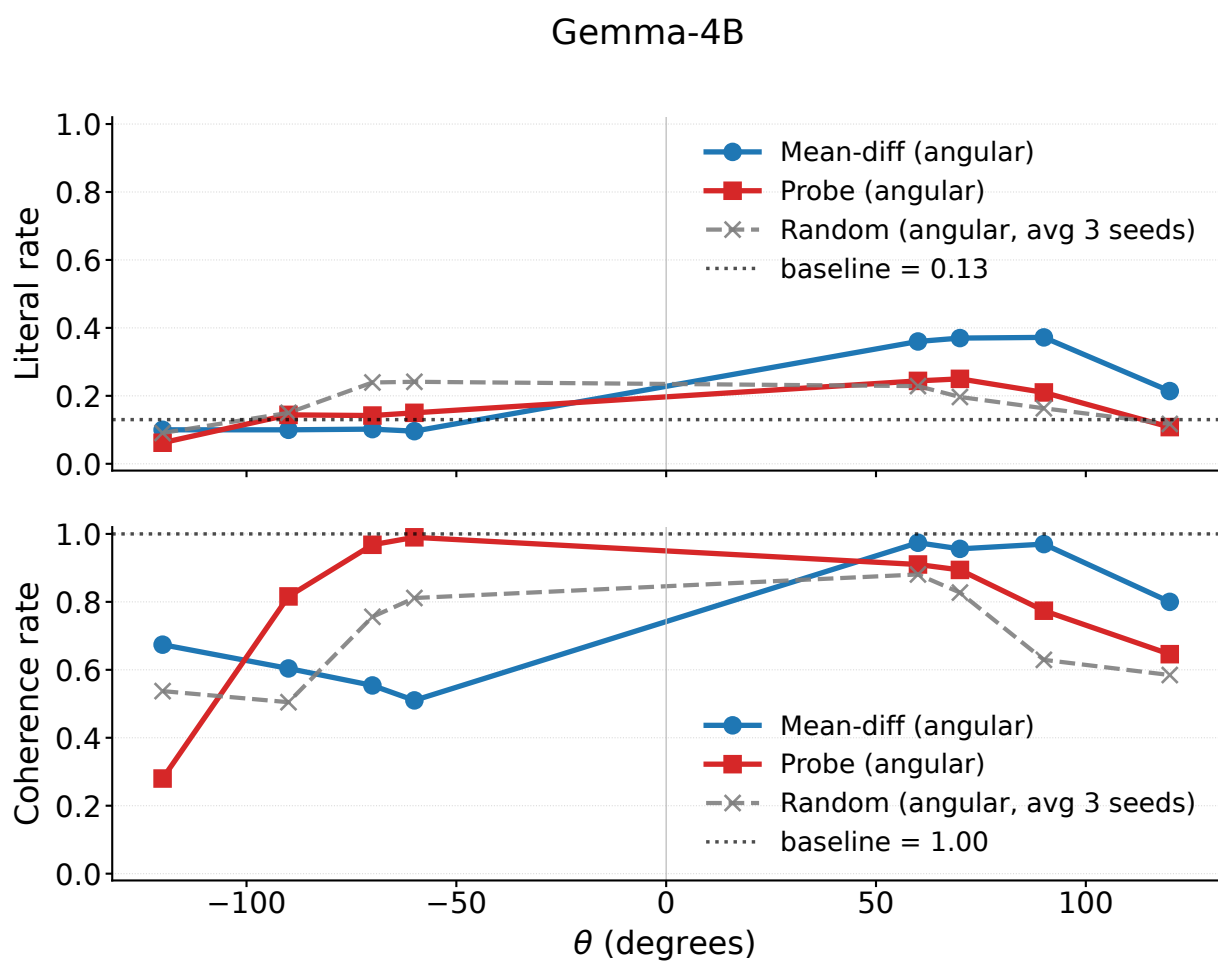


Figure 12: Full angular coefficient sweep on Gemma-3-4B. Top: literal rate vs θ . Bottom: coherence rate.

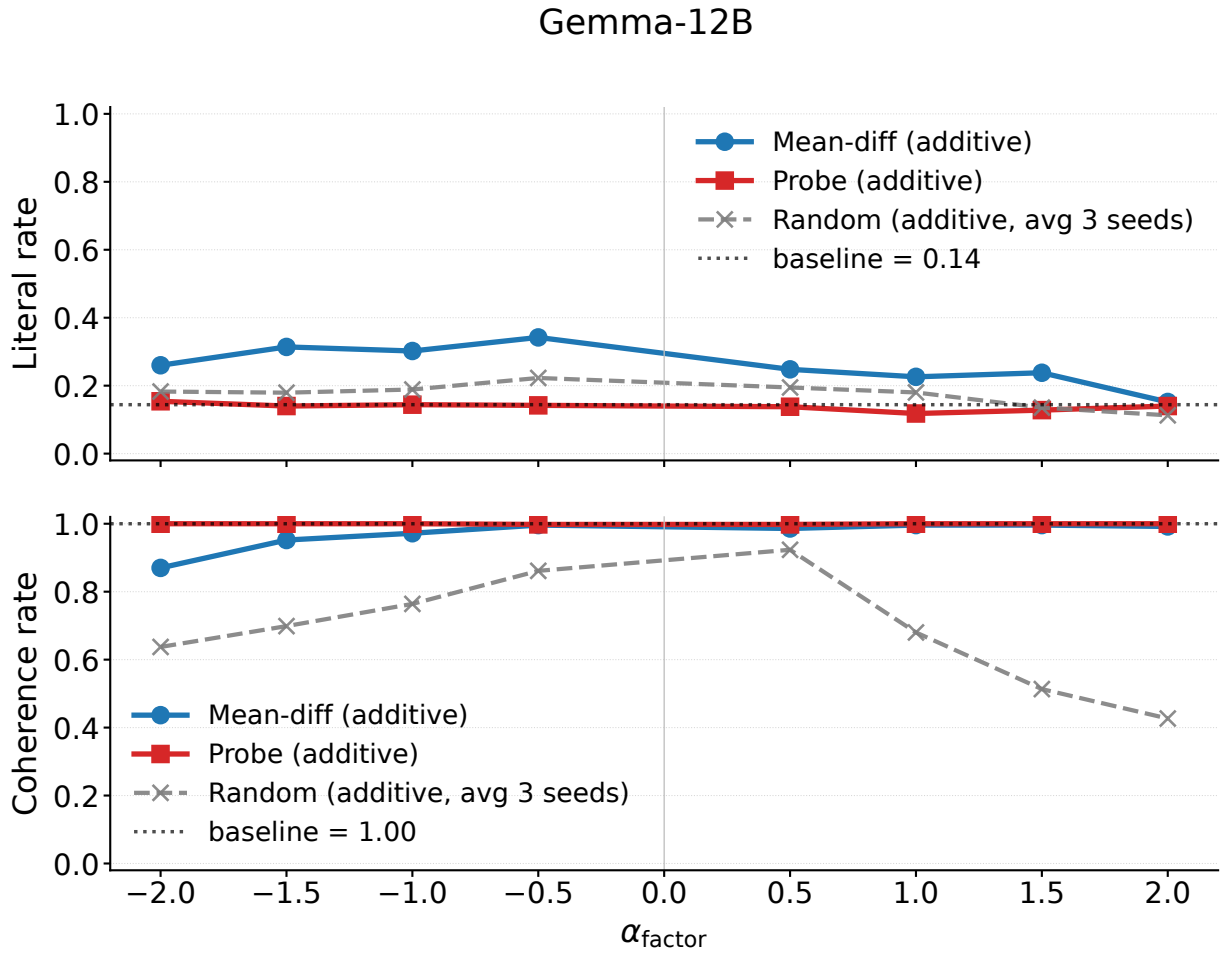


Figure 13: Full additive coefficient sweep on Gemma-3-12B. Top: literal rate vs α_{factor} . Bottom: coherence rate.

Gemma-12B

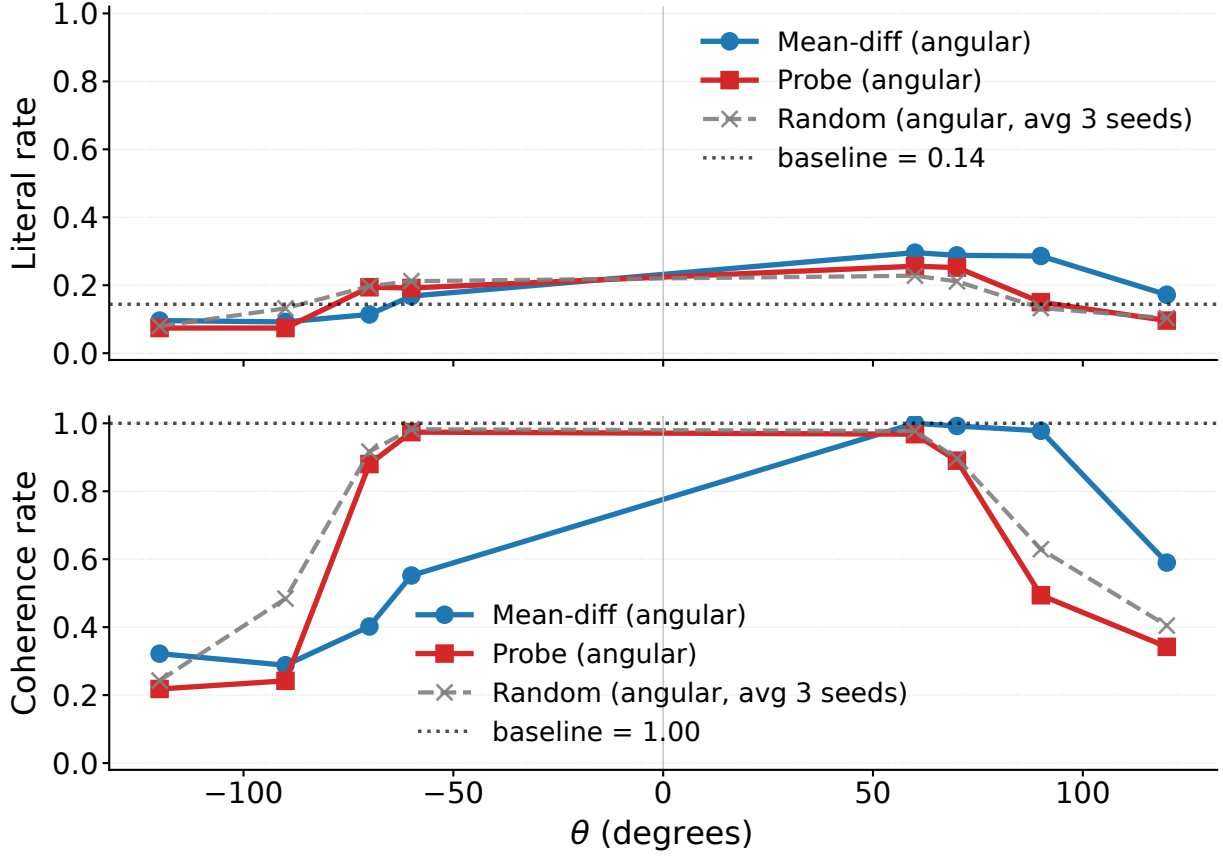


Figure 14: Full angular coefficient sweep on Gemma-12B. Top: literal rate vs θ . Bottom: coherence rate.

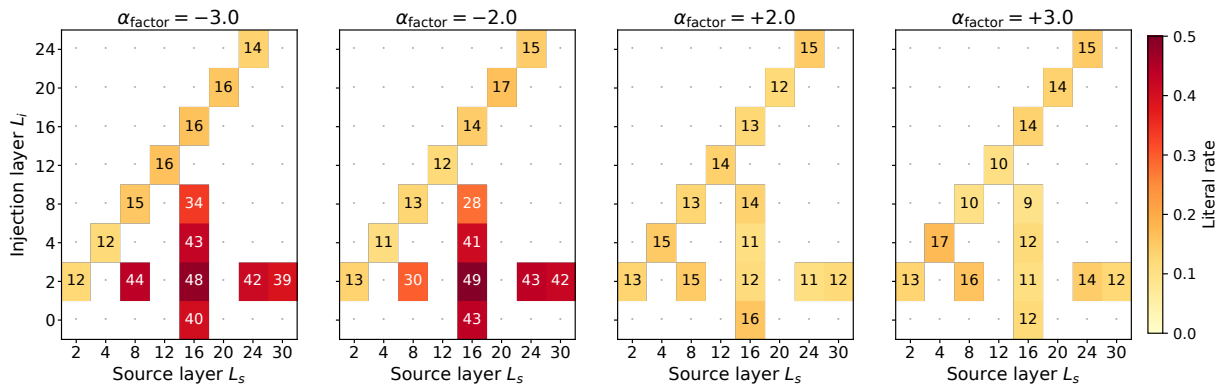


Figure 15: Literal rate over the evaluated (L_s, L_i) grid on Llama-3.1-8B at four multiplier values. Cell labels are literal rates $\times 100$; dotted cells denote layer pairs that were not evaluated. The optimal evaluated region clusters at low L_i and mid-to-late L_s under negative multipliers.

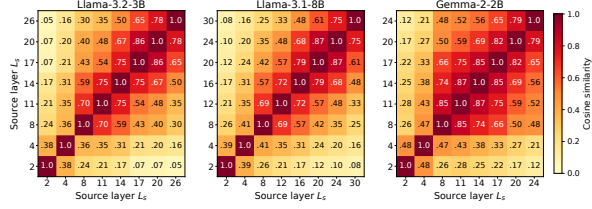


Figure 16: Pairwise cosine similarities between mean-difference steering vectors at different source layers, for three models. A high-cosine cluster in the mid-late layers (cosines 0.67–0.87) indicates directional convergence; lower cosines at the very early or very deep layers indicate distinct directions.

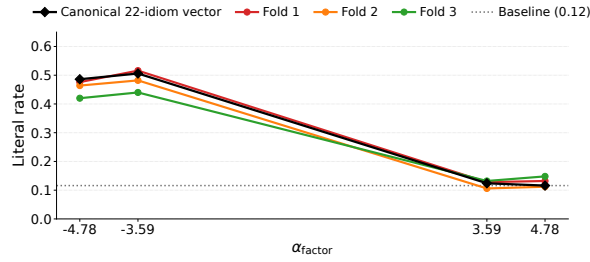


Figure 17: Cross-fold robustness on Llama-3.2-3B: literal rate vs α_{factor} for the canonical 22-idiom vector and three fold-derived vectors trained on disjoint 10–11-idiom subsets, all evaluated at $\alpha_{\text{factor}} \in \{\pm 3.59, \pm 4.78\}$. Fold curves track the canonical curve closely at the tested points.